SET THEORY

ILDOO KIM

ABSTRACT. This lecture note is written to teach undergraduate students a set theory. This lecture note is based on the book [1].

1. Historical Introduction

Paradox 1.1 (Russell's paradox (a logical paradox))**.**
- *If A is a set, its elements may themselves be sets. For instance, consider a set of lines. Then each line itself is a set of points.*
- *There is a possibility that A may be an element of itself, i.e. $A \in A$. For instance, consider the set of all sets. Then obviously, the set of all sets is an element of the set of all sets.*
- *Let S denote* **the set of all sets that are not elements of themselves**. *There are two possibilities that $S \in S$ or $S \notin S$.*
  - *(Case 1) Assume that $S \in S$. Then by the definition of S, S is not an element of S, which is a contradiction to the assumption $S \in S$.*
  - *(Case 2) Assume that $S \notin S$. Then S is a set which is not an element of S itself. Therefore $S \in S$ and get a contradiction.*

Exercise 1.2. *Suggest a way to solve Russell's paradox.*

Russell's paradox naturally leads to the question if the set of all sets that are not elements of themselves exists. To answer the question, we should know what set is first. Here is an another question. "Do we know what set is rigorously?".

Definition 1.3 (Cantor's definition of a set)**.** *Cantor defined a set as "a collection of definite distinguishable objects of our perception which can be conceived as a whole". More specifically, Cantor accepted the "common-sense" notion that "if we can describe a property of objects, then we can also speak of the set of all objects possessing that property."*

Paradox 1.4 (Berry's paradox (a semantic paradox))**.**
- *Let T be the set of all natural numbers that can be described in fewer than twenty words of the English language.*
- *Since there are only a finite number of English words, there are only finitely many combinations of fewer than twenty words. Thus T is a finite set.*
- *Since T is a finite set, there exists a natural number N which is greater than all the elements of T. Thus by the definition of T, N is not an natural number which can be described in fewer than twenty words of the English language.*
- *However, N can be described with seventeen words in the way that "N is an natural number which cannot be described in fewer than twenty words*

of the English language". Therefore by the definition of N, $N \in T$, which leads to the contradiction that $N > N$.

Exercise 1.5. *Suggest a way to solve Berry's paradox.*

One of ways to overcome paradoxes is the axiomatic method and it is called the axiomatic set theory. The axiomatic method in mathematics emerged about 300 B.C. with the appearance of Euclid's Elements (the Euclidean geometry). Before introducing Euclidean geometry, we distinguish an axiom from definitions.

Definition 1.6 (Definition and Axiom)**.**
- *An* **axiom or postulate** *is a statement that is taken to be true.*
- *A* **definition** *is used to give a precise meaning to a new term by describing a condition which unambiguously qualifies what a mathematical term is.*

Recall that Euclid's method consists in assuming a small set of intuitively appealing axioms, and deducing many other propositions (theorems) from these.

Definition 1.7 (Euclidean geometry)**.** *Euclidean geometry is an axiomatic system where all theorems ("true statements") are derived from a small number of simple axioms. We introduce the Euclidean Axioms*

(1) *We can draw a straight line from any point to any point.*
(2) *We can produce an extension of a finite straight line continuously in a straight line.*
(3) *We can draw a circle with any centre and radius.*
(4) *All right angles are equal to one another.*
(5) *(The parallel postulate) If a straight line falling on two straight lines make the interior angles on the same side less than two right angles, the two straight lines meet on that side on which the angles are less than two right angles.*

Here a point and a line are undefined notions even though they are intuitively understandable. Thus we may interpret axioms as properties which some undefined notions naturally have.

Remark 1.8. There appeared non-Euclidean geometries such as hyperbolic geometry and elliptic geometry by replacing the parallel postulate.

The construction of the axiomatic set theory is based on methods of the Euclidean geometric. Thus mathematicians had found axioms with some undefined notions to overcome paradoxes and we will study these axioms and theorems derived from them.

Idea 1.9 (Zermelo's idea)**.**
- *There is one primitive relation denoted by $\in$, that is, $\in$ is a undefined notion and it is understood as a membership relation intuitively.*
- *The expression $x \in Y$ is to be read "x is an element of Y".*
- *The variables x, y, z, X, Y, Z, $\dots$ which are placed to the right or to the left of the symbol $\in$ are called "sets".*
- *Regard elements of every set to be sets themselves. In other words, every set is considered to be a set of sets. Usually, small variables x, y, and z denote elements of sets and capital variables X, Y, and Z indicate sets. However, due to this identification, all expressions such as $x \in Y$, $X \in Y$, $Z \in x$ are meaningful.*

One of the most natural ways to construct sets is considering a collection of objects satisfying a condition. We symbolize this condition by $S(x)$. Then a set is given by a collection of all x satisfying $S(x)$. If we adopt this concept of a set without any restrictions, then we are led to paradoxes. For example, if we define

$$S(x) = \text{not element of itself (or symbolically } x \notin x)$$

and consider the set of all sets x satisfying $S(x)$. Then we fall in Russell's paradox (Paradox 1.1). Therefore, we reach the conclusion that we cannot form the set of all x which satisfy $S(x)$ in general. Zermelo suggested the following axiom.

Axiom 1.10 (Zermelo's axiom of selection)**.** *Let A be a set and $S(x)$ be a statement about x which is meaningful for every x in A. Then there exists a set which consists of exactly those elements which satisfy $S(x)$ in A. It is customarily denoted by*

$$\{x \in A : S(x)\}$$

and it is to be read "the set of all x in A such that $S(x)$".

Idea 1.11 (Zermelo's idea and Russell's paradox)**.** *How does the axiom of selection overcome Russell's paradox? Note that the set of all sets which are not elements of themselves can be symbolized as $\{x : x \notin x\}$. However, the existence of the set $\{x : x \notin x\}$ is not guaranteed by the axiom of selection. The best candidate with the existence for the set of all sets which are not elements of themselves is given by*

$$\{x \in A : x \notin x\},$$

where A is a set which can be shown to exist. Denote $S = \{x \in A : x \notin x\}$. Then Russell's argument is not a paradox anymore. Indeed, assume $S \in S$. Then $S \in S$ implies that $S \notin S$. It is a contradiction. Therefore

$$S \notin S. \tag{1.1}$$

Next we claim that $S \notin A$ and use the reduction to absurdity. Assume that $S \in A$. Then by (1.1)*, we have $S \notin S$ and $S \in A$. By the definition of S, we conclude that $S \in S$, which is contradiction to* (1.1)*. Therefore, we reach the conclusion that $S \notin A$ and there is no paradox. In other words, Russell's argument proves that for any set A, the set $\{x \in A : x \notin x\}$ cannot be an element of A.*

Remark 1.12. Zermelo considered that Russell's set such as the set of all sets that are not elements of themselves is "too large" and excluded too large sets to avoid the paradox.

Question 1.13. *Here is an another natural question. Does Zermelo's idea overcome Berry's paradox (Paradox 1.4)? The answer is "no". Recall that T is the set of all natural numbers that can be described in fewer than twenty words of the English language. Let $\mathbb{N}$ be the set of all natural numbers and*

$$S(x) = x \textit{ can be described in fewer than twenty words of the English language.}$$

Then $T = \{x \in \mathbb{N} : S(x)\}$. Recall that since there are only a finite number of English words, there are only finitely many combinations of fewer than twenty words. Thus there exists a natural number $n \in \mathbb{N}$ such that $n > x$ for all $x \in T$. However, n can be described with seventeen words in the following way:"n is an natural number which cannot be described in fewer than twenty words of the English language". Definitely, it is a paradox. Zermelo did not give a satisfactory answer to solve Berry's paradox but suggested the idea that "we must place restrictions on the type *of "conditions" $S(x)$. That is, the set $\{x \in A : S(x)\}$ can be formed only if $S(x)$ is meaningful for every $x \in A$. He also raised the questions: How do we understand "meaningful"? How do we determine whether $S(x)$ is meaningful?*

Idea 1.14 (Von Neumann's idea)**.**
- *Zermelo used only one undefined notation called the membership relation $x \in Y$. Von Neumann added one more undefined notion called a class which is more general than a set. Von Neumann distinguished between two kinds of classes, namely "elements" and "proper classes". Here "elements" are defined to be those classes which are elements of classes and "proper classes" are defined to be those classes which are not elements of any class.*
- *Zermelo excluded all too large sets. On the other hand, Von Neumann allowed too large sets to exist but they are not permitted to be elements of classes.*
- *Von Neumann suggested an axiom called "the class axiom" which replaces Zermelo's axiom of selection*

Axiom 1.15 (the class axiom)**.** *If $S(x)$ is any statement about an object x, there exists a class which consists of all those elements x which satisfy $S(x)$. In other words, if $S(x)$ is any statement about x, we can form the class*

$$\{x : x \text{ is an element and } S(x)\}.$$

Apply the class axiom to Russell's paradox. Denote

$$S = \{x : x \text{ is an element and } x \notin x\}.$$

Then $S \in S$ is impossible since $S \in S$ implies that $S \notin S$, which is a contradiction. Thus we have

$$S \notin S. \tag{1.2}$$

It follows that S is not an element. Indeed, if S is an element, then we have $S \in S$ by (1.2) and the definition of S, which is a contradiction. Therefore, by the reduction to absurdity, we conclude that S is not an element. There is no paradox in the above arguments. Instead, we reach the conclusion that S is not an element.

Next we want to check if Berry's paradox is solved through the class axiom. According to Zermelo's question, it is possible if we give a restriction on a property $S(x)$. In other words, we need to specify "statements" in the class axiom. Therefore, Von Neumann only admitted "statements" which can be written in **the formal language**. Here "statements" in the formal language mean that all properties in statements can be expressed entirely in terms of the seven mathematical symbols $\in$, $\vee$, $\wedge$, $\neg$, $\implies$, $\forall$, $\exists$ and variables x, y, z, $\ldots$.

Remark 1.16. We will study an axiomatic theory based on Von Neumann's idea in this lecture. You can determine by yourself if you accept axioms or not. Sometimes, it could make a meaningful progress by denying axioms. For instance, denying the parallel postulate in Euclidean geometry initiated developments of non-Euclidean geometries. However, keep in mind that most modern mathematical theories (especially analysis theories) are based on axiomatic set theories. Hilbert once wrote "We will not be expelled from the paradise into which Cantor has led us".

2. Sentence and Logic

Definition 2.1 (Sentence)**.** *"Sentence" is a statement which is unambiguously either true or false (in a given context). For example,*

Seoul is the capital of Korea,

Money grows on trees,

and

Snow is black

are all sentences.

Exercise 2.2. *Is "Korea university is the top university in Korea" a sentence?*

We will use letters P, Q, R, S to denote sentences.
By using a given sentence, we can produce a new sentence.

Definition 2.3 (Negation)**.** *Let P be a sentence. Then the negation of P is understood to assert that "P is false". In other words, if P is true, then the negation of P is false. The negation of P is denoted by $\neg P$ and it is read as "not P". We use "t" and "f" to denote "true" and "false", respectively in the following table.*

Truth table for the negation	
P	$\neg P$
t	f
f	t

By combining sentences, we can make a new sentence. Here is a list of combinations of sentences.

Definition 2.4 (Conjunction)**.** *Let P and Q be sentences. Then the conjunction of P and Q is understood to assert that "the conjunction of P and Q is true if P is true and Q is true". In other words, if both P and Q are true, then the conjunction of P and Q is true and otherwise it is false. The conjunction of P and Q is denoted by $P \wedge Q$ and it is read as "P and Q".*

Truth table for the conjunction		
P	Q	$P \wedge Q$
t	t	t
t	f	f
f	t	f
f	f	f

Definition 2.5 (Disjunction)**.** *Let P and Q be sentences. Then the disjunction of P and Q is understood to assert that "the disjunction of P and Q is true if P, or Q, or both P and Q are true". In other words, the disjunction of P and Q is false only if both P and Q are false. The disjunction of P and Q is denoted by $P \vee Q$ and it is read as "P or Q".*

Truth table for the disjunction		
P	Q	$P \vee Q$
t	t	t
t	f	t
f	t	t
f	f	f

Definition 2.6 (Implication)**.** *Let P and Q be sentences. Then that P implies Q is understood to assert that "if P is true, then Q is true". More specifically in mathematics, the implication is true except if P is true and Q is false. We use the notation $P \implies Q$ to denote that P implies Q.*

Truth table for the implication		
P	Q	$P \implies Q$
t	t	t
t	f	f
f	t	t
f	f	t

Notation 2.7.
- *By the definition, $P \implies Q$ is read as "P implies Q".*
- *$P \implies Q$ is also read as "if P, then Q" or "Q if P".*

Example 2.8.
(1) The truth table for the implication is working even though there is no relation between two sentences. For instance, the sentence

$$1 + 1 = 2 \quad \implies \quad \pi \text{ is an irrational number}$$

is true since both sentences "$1 + 1 = 2$" and "π is an irrational number" are true.

(2) If the premise P is true, then the truth of the implication $P \implies Q$ is determined by truth of Q and it seems very natural to our common sense. However, it is difficult to understand the implication if the premise P is false. Note that if the premise P is false, then the implication $P \implies Q$ is always true due to our truth table. Consider the following example:

$$1 + 1 = 3 \quad \implies \quad 3 \text{ is an even number.}$$

Both sentences above are false but the implication is true by our truth rule.

(3) Let A be the empty set, i.e. A has no elements. Consider the sentence

$$\text{for all } x \in A, \ x \text{ can have any property.}$$

We could interpret the above sentence by using the implication notation as follows:

$$x \in A \implies x \text{ can have any property.}$$

Therefore, this sentence is true. In other words, if the given set A is empty, then we may say that all elements of A have any property which we want to give. This kind of situation appears frequently in mathematics when we consider the extreme case that the given set is empty.

(4) Let P be a sentence. Then $P \implies P$ is always true.

Exercise 2.9. *Let $\mathbb{N}$ be the natural number system. Determine if the following two sentence are true or false:*

(1)

$$A \subset \mathbb{N} \implies \text{all elements of } A \text{ are integers}$$

(2)

$$A \subset \mathbb{N} \implies \text{all elements of } A \text{ are even numbers.}$$

Definition 2.10 (The converse, inverse, contrapositive)**.** *Let P and Q be sentences and consider the implication that P implies Q, i.e. $P \implies Q$.*

(1) The **converse** *of the implication is that Q implies P, i.e.*

$$Q \implies P.$$

(2) The **inverse** *of the implication is that $\neg P$ implies $\neg Q$, i.e.*

$$\neg P \implies \neg Q.$$

(3) The **contrapositive** *of the implication is that $\neg Q$ implies $\neg P$, i.e.*

$$\neg Q \implies \neg P.$$

Remark 2.11. Let P and Q be sentences and consider the implication that P implies Q. The truth tables for the converse, inverse, and contrapositive are easily obtained from truth values from the negation and implication. Indeed,

Truth table for the converse of the implication		
P	Q	$Q \implies P$
t	t	t
t	f	t
f	t	f
f	f	t

Truth table for the inverse of the implication				
P	Q	$\neg P$	$\neg Q$	$\neg P \implies \neg Q$
t	t	f	f	t
t	f	f	t	t
f	t	t	f	f
f	f	t	t	t

Truth table for the contrapositive of the implication				
P	Q	$\neg P$	$\neg Q$	$\neg Q \implies \neg P$
t	t	f	f	t
t	f	f	t	f
f	t	t	f	t
f	f	t	t	t

We emphasize that the truth values for $P \implies Q$ and its contrapositive are same.

Notation 2.12. *$P \implies Q$ is also read as "P only if Q", "P is a sufficient condition of Q", and "Q is a necessary condition of P".*

Theorem 2.13. *Let P and Q be sentences. Then the following statements are (always) true:*

(1)
$$P \implies P \vee Q$$

(2)
$$Q \implies P \vee Q$$

(3)
$$P \wedge Q \implies P$$

(4)
$$P \wedge Q \implies Q$$

Proof. (1) Recall the truth table for the conjunction and implication. We divide the proof into four cases.

- (Case 1: P is true and Q is true). If P is true and Q is true, then $P \vee Q$ is true. Thus $P \implies P \vee Q$ is true.
- (Case 2: P is true and Q is false). If P is true and Q is false, then $P \vee Q$ is true. Thus $P \implies P \vee Q$ is true.
- (Case 3: P is false and Q is true). If P is false, then $P \implies P \vee Q$ is true.
- (Case 4: P is false and Q is false). If P is false, then $P \implies P \vee Q$ is true.

Therefore

$$P \implies P \vee Q$$

We can also deduce the conclusion through the following truth table:

Truth table for $P \implies P \vee Q$			
P	Q	$P \vee Q$	$P \implies P \vee Q$
t	t	t	t
t	f	t	t
f	t	t	t
f	f	f	t

(2) The truth table for $Q \implies P \vee Q$ is analogous to the one for $P \implies P \vee Q$. We skip the detail.

(3) We derive a truth table for $P \wedge Q \implies P$.

Truth table for $P \wedge Q \implies P$			
P	Q	$P \wedge Q$	$P \wedge Q \implies P$
t	t	t	t
t	f	f	t
f	t	f	t
f	f	f	t

(4) Since the truth table for $P \wedge Q \implies Q$ is very similar to the one for $P \wedge Q \implies P$, we skip the detail.

□

Theorem 2.14. *Let P, Q, and R be sentences. Assume that $Q \implies R$ is true. Then the following statements are true:*

(1)

$$P \vee Q \implies P \vee R$$

(2)

$$P \wedge Q \implies P \wedge R.$$

Proof. (1) We consider the truth table for $P \vee Q \implies P \vee R$ first.

Truth table for $P \vee Q \implies P \vee R$					
P	Q	R	$P \vee Q$	$P \vee R$	$P \vee Q \implies P \vee R$
t	t	t	t	t	t
t	t	f	t	t	t
t	f	t	t	t	t
t	f	f	t	t	t
f	t	t	t	t	t
f	t	f	t	f	f
f	f	t	f	t	t
f	f	f	f	f	t

Recalling the assumption that $Q \implies R$ is true, we consider the truth table for $Q \implies R$.

Truth table for $Q \implies R$		
Q	R	$Q \implies R$
t	t	t
t	f	f
f	t	t
f	f	t

Due to the assumption, the case that Q is true and R is false is impossible. Remove this case from the truth table for $P \vee Q \implies P \vee R$. Then we have

Under the assumption that $Q \implies R$					
P	Q	R	$P \vee Q$	$P \vee R$	$P \vee Q \implies P \vee R$
t	t	t	t	t	t
t	f	t	t	t	t
t	f	f	t	t	t
f	t	t	t	t	t
f	f	t	f	t	t
f	f	f	f	f	t

Therefore, we conclude that $P \vee Q \implies P \vee R$ is always true under the assumption that $Q \implies R$.

(2) The proof of (2) is analogous to (1). We leave the proof as an exercise.

□

Remark 2.15. In Theorem 2.14, we assumed that $Q \implies R$ is true. However, conventionally, we usually skip "is true". In other words, if we simply say that $Q \implies R$, then it means that the sentence $Q \implies R$ is true.

Exercise 2.16. *Prove (2) in Theorem 2.14.*

Theorem 2.17 (Transitive law)**.** *Let P, Q, and R be sentences. Then*

$$[(P \implies Q) \wedge (Q \implies R)] \implies (P \implies R).$$

Proof. If $(P \implies Q) \wedge (Q \implies R)$ is false, then the implication is true. Thus we only need to consider the case $(P \implies Q) \wedge (Q \implies R)$ is true and by the truth table for the conjunction, we may assume that both $P \implies Q$ and $Q \implies R$ are true. Let's classify all cases that both $P \implies Q$ and $Q \implies R$ are true. Consider the truth table for $P \implies Q$ and $Q \implies R$.

Truth table for $P \implies Q$		
P	Q	$P \implies Q$
t	t	t
t	f	f
f	t	t
f	f	t

and

Truth table for $Q \implies R$		
Q	R	$Q \implies R$
t	t	t
t	f	f
f	t	t
f	f	t

Then all possible cases are

Under the assumption that both $P \implies Q$ and $Q \implies R$ are true		
P	Q	R
t	t	t
f	t	t
f	f	t
f	f	f

and we have

Under the assumption that both $P \implies Q$ and $Q \implies R$ are true		
P	R	$P \implies R$
t	t	t
f	t	t
f	t	t
f	f	t

The theorem is proved. $\square$

Notation 2.18. *We use the abbreviation $P \iff Q$ for $(P \implies Q) \wedge (Q \implies P)$. $P \iff Q$ is read as "P if and only if Q". We also say that two sentences P and Q are "equivalent".*

Theorem 2.19. *Let P, Q, and R be sentences. Then*

(1) *(Idempotent law)*
$$P \vee P \iff P$$
(2) *(Idempotent law)*
$$P \wedge P \iff P$$
(3) *(Commutative law)*
$$P \vee Q \iff Q \vee P$$
(4) *(Commutative law)*
$$P \wedge Q \iff Q \wedge P$$

(5) *(Associative law)*
$$P \vee (Q \vee R) \iff (P \vee Q) \vee R$$
(6) *(Associative law)*
$$P \wedge (Q \wedge R) \iff (P \wedge Q) \wedge R$$
(7) *(Distributive law)*
$$P \wedge (Q \vee R) \iff (P \wedge Q) \vee (P \wedge R)$$
(8) *(Distributive law)*
$$P \vee (Q \wedge R) \iff (P \vee Q) \wedge (P \vee R).$$

Proof. (1) The "if part" is obvious. Indeed, if P is true, then P is true or P is true. Therefore $P \vee P$ is true. Next we prove the "only if part". If $P \vee P$ is false, then there is nothing to prove. Thus it is sufficient to show that if $P \vee P$ is true, then P is true. Assume that $P \vee P$ is true. Then by the truth table for the disjunction, P is true or P is true. Therefore P is true.

We leave the proof of the other statements as exercises. □

Exercise 2.20. *Show that for every sentence P,*
$$P \iff \neg(\neg P).$$

Exercise 2.21. *Let P and Q be sentences. Show that*
$$\neg(P \vee Q) \iff \neg P \wedge \neg Q$$
and
$$\neg(P \wedge Q) \iff \neg P \vee \neg Q.$$

Finally we close this section by introducing math symbols which will be used in this lecture note and overall mathematics.

Notation 2.22.
- *We use the abbreviation "iff" to denote "if and only if", i.e. iff means $\iff$.*
- *$\wedge$ is used for "and".*
- *$\vee$ is used for "or".*
- *For all sentences, P, Q, R, ...,*
$$P \implies Q \implies R \implies \cdots$$
should be understood to mean that $P \implies Q$, $Q \implies R$, and so on.
- *$\exists$ is used to denote "there exists"*
- *$\forall$ means "for all".*
- *$\ni$ is read as "such that" and it is used to describe properties.*

3. Classes and Sets

Undefined notion 3.1 (Class and $\in$)**.** (1) *"Class" is an undefined notion.*
(2) *"Membership relation" denoted by $\in$ is an undefined notion.*

Remark 3.2. Intuitively, a class can be understood as a collection of objects satisfying a certain property.

Definition 3.3.
- *For a class x, we say that x is a "set" or "element" if there exists a class A such that $x \in A$.*

- *More specifically, we say that x is an element of A if $x \in A$.*
- *The expression $x \in A$ is read as "x is an element of A", or "x belongs to A", or simply "x is in A.*
- *Moreover, we write $x \notin A$ for "x is not an element of A", i.e.*

$$x \notin A \iff \neg(x \in A).$$

- *A class X is called a "proper class" if X is not an element, i.e. $X \notin A$ for all class A.*

Remark 3.4. Logically, it seems to be more correct that a class x is called a "set" iff (if and only if) there exists a class A such that $x \in A$. However, for definitions, conventionally we simply say that a class x is called a "set" if there exists a class A such that $x \in A$.

Notation 3.5.
- *Small letters a, b, c, x, y, z, ... will be used only to designate elements. In other words, if we use small letters a, b, c, x, y, z, ..., then they are all elements.*
- *Capital letters A, B, C, X, Y, Z, ... may denote a (general) class. In other words, capital letters A, B, C, X, Y, Z, ... can be either an element or a proper class.*

Definition 3.6. *Let A and B classes. We define $A = B$ to mean that every class that has A as an element also has B as an element, and vice-versa. In symbols,*

$$A = B \quad \textit{iff} \quad \forall X,\ A \in X \iff B \in X.$$

In other words, two classes to be equal if and only if they are members of the same classes. Moreover, we use the notation $A \neq B$ to denote $\neg(A = B)$, i.e.

$$A \neq B \quad \textit{iff} \quad \exists X \textit{ such that } A \in X \ \wedge\ B \notin X \quad \textit{or} \quad A \notin X \ \wedge\ B \in X.$$

Exercise 3.7. *Let A and B be proper classes. Prove that $A = B$.*

Remark 3.8.
- The above exercise tells us that Definition 3.6 does not give any distinction among proper classes.
- Note that any proper class cannot be an element but it can contain elements i.e. for a proper class A, it is possible that there exists an element $x \in A$.

Axiom 3.9 (Axiom of extent)**.** *For all classes A and B,*

$$A = B \quad \textit{iff} \quad x \in A \iff x \in B.$$

Exercise 3.10. *Let A and B classes. Explain the meaning of $A \neq B$ on the basis of the axiom of extent.*

Definition 3.11. *Let A and B classes.*

- *We define $A \subset B$ to mean that every element of A is an element of B, i.e.*

$$A \subset B \quad \textit{iff} \quad x \in A \implies x \in B.$$

- *If $A \subset B$, then we say that A is a "subclass" of B.*
- *We define $A \subsetneq B$ to mean that $A \subset B$ and $A \neq B$. In this case, we say that A is a "strict subclass" of B.*
- *If A is a subclass of B and A is a set, then we call A a "subset" of B. Moreover, if A is a strict subclass and a set, then A is called a "proper subset".*

- *By the axiom of extent,*
$$A \subset B \quad \textit{and} \quad B \subset A \iff A = B.$$

Remark 3.12. Don't be confused with the notation in our book. In the book, $A \subseteq B$ and $A \subset B$ are used to denote that A is a subclass of B and A is a strict subclass of B, respectively. I changed the notation since I believe that $A \subset B$ and $A \subsetneq B$ are used more frequently in mathematics.

Theorem 3.13. *Let A, B, and C be classes. Then*

(1)
$$A = A.$$

(2)
$$A = B \implies B = A.$$

(3)
$$A = B \textit{ and } B = C \implies A = C.$$

(4)
$$A \subset B \textit{ and } B \subset A \implies A = B.$$

(5)
$$A \subset B \textit{ and } B \subset C \implies A \subset C.$$

Proof. We use the axiom of extent to prove these statements.

(1) Obviously
$$x \in A \iff x \in A.$$
Thus by the axiom of extent, $A = A$.

(2) Suppose that $A = B$. Then by the axiom of extent,
$$x \in A \iff x \in B.$$
Thus by the definition of $\iff$ and Theorem 2.19(4),
$$x \in B \iff x \in A.$$
Finally by the axiom of extent, we have $B = A$.

(3) Suppose that $A = B$ and $B = C$. Then we have the following four statements:
$$x \in A \implies x \in B \tag{3.1}$$
$$x \in B \implies x \in A \tag{3.2}$$
$$x \in B \implies x \in C \tag{3.3}$$
$$x \in C \implies x \in B. \tag{3.4}$$
By (3.1), (3.3), and Theorem 2.17,
$$x \in A \implies x \in C. \tag{3.5}$$
Similarly, (3.4), (3.2), and Theorem 2.17,
$$x \in C \implies x \in A. \tag{3.6}$$
Combining (3.5) and (3.6), by the axiom of extent, we have
$$A = C.$$
We leave the proof of (4) and (5) as exercises.

$\square$

Exercise 3.14. *Prove (4) and (5) in Theorem 3.13.*

Exercise 3.15. *Prove (1) - (3) in Theorem 3.13 without using the axiom of extent.*

Axiom 3.16 (Axiom of class construction)**.** *Let $P(x)$ designate a statement about x in the formal language, i.e. $P(x)$ can be expressed entirely in terms of seven symbols $\in$, $\vee$, $\wedge$, $\neg$, $\implies$, $\exists$, $\forall$, brackets, and variables x, y, z, A, B, ... Then there exists a class C which consists of all the elements x which satisfy $P(x)$. In other words, the class $\{x : x \text{ is an element and } P(x)\}$ exists.*

Remark 3.17.
(1) Recall that the notation $\ni$ means "such that" and it will be contained in the formal language since $\ni$ and $\in$ symmetric.
(2) Note that the axiom of class construction only permits us to form the class of all the elements x which satisfy $P(x)$. Thus, the axiom does not guarantee the existence of the class of all the classes x which satisfy $P(x)$. Considering only elements is necessary to avoid logical paradoxes.
(3) Recall that small letters x, y, z, ... only denote elements. Thus simly we can consider the class of the form of $\{x : P(x)\}$ and it exists by the axiom of class construction.

Exercise 3.18. *Explain why the class of all the classes X which satisfy $P(X)$ cannot exists in general.*

Definition 3.19. *Let A and B be classes. The "intersection" of A and B is defined to be the class of all the elements which belong to both A and B. In symbols,*

$$A \cap B := \{x : x \in A \textit{ and } x \in B\},$$

where we used the notation $:=$ instead of $=$ to emphasize that this equality is a definition.

Remark 3.20. Note that the symbol $\wedge$ means "and". Thus

$$\{x : x \in A \text{ and } x \in B\} = \{x : x \in A \ \wedge \ x \in B\}.$$

Therefore, the class $A \cap B$ exists by the axiom of class construction.

Definition 3.21. *Let A and B be classes. The "union" of A and B is defined to be the class of all the elements which belong either to A, or B, or to both A and B. In symbols,*

$$A \cup B := \{x : x \in A \textit{ or } x \in B\},$$

Exercise 3.22. *Prove that for all classes A and B, the union $A \cup B$ exists.*

Definition 3.23. *The "universal class" is defined to be the class of all elements and it is denoted by $\mathcal{U}$. By Theorem 3.13(1), every element x satisfies $x = x$. Thus by using mathematical symbols, we can write*

$$\mathcal{U} = \{x : x = x\}.$$

Therefore the existence of the universal class $\mathcal{U}$ is guaranteed by the axiom of class construction.

Definition 3.24. *The "empty class" is defined to be the class which has no elements at all and it is denoted by $\emptyset$. By Theorem 3.13(1), every element x satisfies $x = x$. In other words, there is no element x such that $x \neq x$. Therefore we can write the empty class in symbols as follows:*

$$\emptyset = \{x : x \neq x\}.$$

The existence of the empty class $\emptyset$ is guaranteed by the axiom of class construction

Theorem 3.25. *For every class A,*

(1)

$$\emptyset \subset A$$

(2)

$$A \subset \mathcal{U}.$$

Proof. (1) It is sufficient to prove that $x \in \emptyset \implies x \in A$ by the definition of $\subset$. We prove the contrapositive of the statement, i.e. we prove $x \notin A \implies x \notin \emptyset$. Suppose that $x \notin A$. Then obviously $x \notin \emptyset$ since $\emptyset$ has no elements.

(2) Assume $x \in A$. Then x is an element. Thus $x \in \mathcal{U}$.

□

Definition 3.26. *If two classes have no elements in common, they are said to be "disjoint". In symbols,*

$$A \text{ and } B \text{ are disjoint} \quad \text{iff} \quad A \cap B = \emptyset.$$

Definition 3.27. *Let A be a class. The "complement" of the class A is the class of all the elements which do not belong to A. In symbols,*

$$A^c = \{x : x \notin A\}.$$

Thus $x \in A^c$ if and only if $x \notin A$.

Relations among classes can be represented graphically by means of a useful device known as the "Venn diagram". For examples, see figures 1-4.

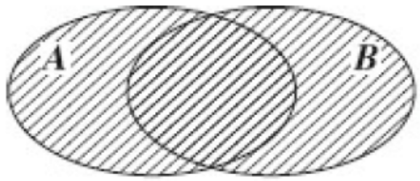

FIGURE 1. $A \cup B$

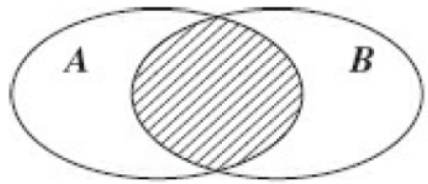

FIGURE 2. $A \cap B$

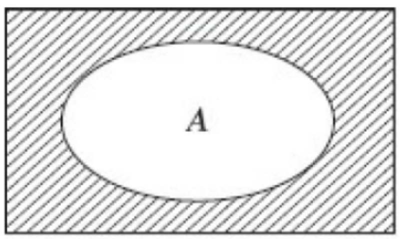

FIGURE 3. A^c

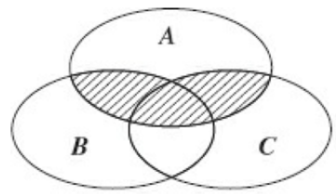

FIGURE 4. $A \cap (B \cup C)$ or $(A \cap B) \cup (A \cap C)$

The Venn diagram does not give rigorous proofs. However, it can be useful since it can give intuitions about relations. For instance, Figure 4 shows us that

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

Theorem 3.28. *Let A and B classes. Then*

(1) $A \subset A \cup B$
(2) $B \subset A \cup B$
(3) $A \cap B \subset A$
(4) $B \cap B \subset B$.

Proof. (1) To prove (1), we must show that $x \in A \implies x \in A \cup B$. By Theorem 2.13(1) and the definition of $A \cup B$,

$$\begin{aligned} x \in A &\implies x \in A \ \vee \ x \in B \\ &\implies x \in A \cup B. \end{aligned}$$

(2) The proof of (2) is almost analogous to (1). We skip the detail.
(3) By the definition of $A \cap B$ and Theorem 2.13(3),

$$\begin{aligned} x \in A \cap B &\implies x \in A \ \wedge \ x \in B \\ &\implies x \in A. \end{aligned}$$

(4) Note that (3) holds for all classes A and B. In particular, (3) holds for all classes $A = B$. Therefore (4) comes from (3).

$\square$

Theorem 3.29. *Let A and B classes. Then*

(1) $A \subset B$ *if and only if* $A \cup B = B$
(2) $A \subset B$ *if and only if* $A \cap B = A$.

Proof. (1) First we prove the only if part. Assume $A \subset B$, i.e.

$$x \in A \implies x \in B. \tag{3.7}$$

Then by the definition of the union, (3.7), Theorem 2.14(1), and Theorem 2.19(1),

$$\begin{aligned} x \in A \cup B &\implies x \in A \quad \text{or} \quad x \in B \\ &\implies x \in B \quad \text{or} \quad x \in B \\ &\implies x \in B. \end{aligned}$$

Thus $A \cup B \subset B$. Moreover, $B \subset A \cup B$ by Theorem 3.28(2). Therefore, $A \cup B = B$. Next we prove the if part. Assume that $A \cup B = B$. Then by 3.28(1),

$$A \subset A \cup B = B.$$

(2) We leave the proof of (2) as an exercise.

□

Exercise 3.30. *Prove Theorem 3.29(2).*

Lemma 3.31. *Let A, B, and C be classes. Assume that $A \subset B$. Then*

(1)
$$A \cup C \subset B \cup C$$

(2)
$$C \cup A \subset C \cup B$$

(3)
$$A \cap C \subset B \cap C$$

(4)
$$C \cap A \subset C \cap B.$$

Proof. Because of similarity of the proof, we only prove (1). By the assumption that $A \subset B$,

$$x \in A \implies x \in B \tag{3.8}$$

Thus by the definition of the union, (3.8), Theorem 2.14(1), and Theorem 2.19(3),

$$\begin{aligned} x \in C \cup A &\implies x \in C \quad \vee \quad x \in A \\ &\implies x \in C \quad \vee \quad x \in B \\ &\implies x \in C \cup B. \end{aligned}$$

□

Theorem 3.32 (Absorption Laws)**.** *Let A and B classes. Then*

(1)
$$A \cup (A \cap B) = A$$

(2)
$$A \cap (A \cup B) = A.$$

Proof. (1) By Theorem 3.28(2),

$$A \subset A \cup (A \cap B) \tag{3.9}$$

On the other hand, by Theorem 3.28(3),

$$A \cap B \subset A.$$

Thus by Lemma 3.31(2), the definition of the union, and Theorem 2.19(1),

$$A \cup (A \cap B) \subset A \cup A = A. \tag{3.10}$$

Combining (3.9) and (3.10), we have

$$A \cup (A \cap B) = A.$$

(2) By Theorem 3.28(3),

$$A \cap (A \cup B) \subset A \tag{3.11}$$

On the other hand, by Theorem 3.28(1),

$$A \subset A \cup B$$

Moreover, by the definition of the intersection, Theorem 2.19(2), and Lemma 3.31(4),

$$A = A \cap A \subset A \cap (A \cup B). \tag{3.12}$$

Combining (3.11) and (3.12), we have

$$A \cap (A \cup B) = A.$$

□

Theorem 3.33. *For every class A,*

$$(A^c)^c = A.$$

Proof. By the axiom of extent, the definition of the complement and $\notin$, and Exercise 2.20,

$$\begin{aligned} x \in (A^c)^c \iff x \notin A^c \iff \neg(x \in A^c) \iff \neg(x \notin A) &\iff \neg(\neg(x \in A)) \\ &\iff x \in A. \end{aligned}$$

□

Theorem 3.34 (DeMorgan's Laws)**.** *Let A and B classes. Then*

(1)

$$(A \cup B)^c = A^c \cap B^c$$

(2)

$$(A \cap B)^c = A^c \cup B^c.$$

Proof. (1) By the axiom of extent, the definitions of the complement, the union, and the intersection, Exercise 2.21,

$$\begin{aligned} x \in (A \cup B)^c \iff x \notin A \cup B \iff \neg(x \in A \cup B) &\iff \neg(x \in A \ \vee \ x \in B) \\ &\iff \neg(x \in A) \ \wedge \ \neg(x \in B) \\ &\iff x \notin A \ \wedge \ x \notin B \\ &\iff x \in A^c \ \wedge \ x \in B^c \\ &\iff x \in A^c \cap B^c. \end{aligned}$$

(2) The proof of (2) is left as an exercise.

□

Exercise 3.35. *Prove (2) in Theorem 3.34.*

Theorem 3.36. *Let A, B, and C be classes. Then*

(1) (Commutative law)

$$A \cup B = B \cup A$$

(2) (Commutative law)

$$A \cap B = B \cap A$$

(3) (Idempotent law)

$$A \cup A = A$$

(4) (Idempotent law)

$$A \cap A = A$$

(5) (Associative law)

$$A \cup (B \cup C) = (A \cup B) \cup C$$

(6) (Associative law)

$$A \cap (B \cap C) = (A \cap B) \cap C$$

(7) (Distributive law)

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

(8) (Distributive law)

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Proof. (1) By the definition of the union and Theorem 2.19(3),

$$\begin{aligned} x \in A \cup B &\iff x \in A \ \vee \ x \in B \\ &\iff x \in B \ \vee \ x \in A \\ &\iff x \in B \cup A. \end{aligned}$$

(2) The proof of (2) is left as an exercise.
(3) The proof of (3) is left as an exercise.
(4) The proof of (4) is left as an exercise.

(5) By the definition of the union and Theorem 2.19(5),

$$\begin{aligned} x \in A \cup (B \cup C) &\iff x \in A \ \vee \ x \in B \cup C \\ &\iff x \in A \ \vee \ (x \in B \ \vee \ x \in C) \\ &\iff (x \in A \ \vee \ x \in B) \ \vee \ x \in C \\ &\iff (x \in A \cup B) \ \vee \ x \in C \\ &\iff x \in (A \cup B) \cup C. \end{aligned}$$

(6) The proof of (6) is left as an exercise.

(7) By the definition of the intersection, and the definition of the union, and Theorem 2.19(7),

$$\begin{aligned} x \in A \cap (B \cup C) &\iff x \in A \ \wedge \ x \in (B \cup C) \\ &\iff x \in A \ \wedge \ (x \in B \ \vee \ x \in C) \\ &\iff (x \in A \ \wedge \ x \in B) \ \vee \ (x \in A \ \wedge \ x \in C) \\ &\iff x \in A \cap B \ \vee \ x \in A \cap C \\ &\iff x \in (A \cap B) \cup (A \cap C). \end{aligned}$$

(8) The proof of (8) is left as an exercise.

□

Exercise 3.37. *Prove (2), (3), (4), (6), and (8) in Theorem 3.36.*

Theorem 3.38. *For every class A,*

(1)

$$A \cup \emptyset = A$$

(2)

$$A \cap \emptyset = \emptyset$$

(3)

$$A \cup \mathcal{U} = \mathcal{U}$$

(4)

$$A \cap \mathcal{U} = A$$

(5)

$$\mathcal{U}^c = \emptyset$$

(6)

$$\emptyset^c = \mathcal{U}$$

(7)

$$A \cup A^c = \mathcal{U}$$

(8)

$$A \cap A^c = \emptyset.$$

Proof. (1) Recalling Theorem 3.25(1), we have

$$\emptyset \subset A.$$

Thus by Theorem 3.29(1),

$$\emptyset \cup A = A.$$

Finally, by the commutative law, we have

$$A \cup \emptyset = A.$$

(2) By Theorem 3.25(1), Theorem 3.29(2), and the commutative law, we have

$$A \cap \emptyset = \emptyset.$$

(3) By Theorem 3.25(2),

$$A \subset \mathcal{U}.$$

Therefore by Theorem 3.29(1),

$$A \cup \mathcal{U} = \mathcal{U}.$$

The proofs of the remaining parts are left as exercises. □

Exercise 3.39. *Prove (4)-(8) in Theorem 3.38.*

By applying theorems introduced in this section, we can prove lots of statements. Here are some examples.

Example 3.40. *Prove that*

$$A \cap (A^c \cup B) = A \cap B.$$

Proof. By the distributive law, Theorem 3.38(8), the commutative law, and Theorem 3.38(2),

$$\begin{aligned} A \cap (A^c \cup B) &= (A \cap A^c) \cup (A \cap B) \\ &= \emptyset \cup (A \cap B) \\ &= (A \cap B) \cup \emptyset \\ &= A \cap B. \end{aligned}$$

□

Definition 3.41. *Let A and B classes. The "difference" of two classes A and B is the class of all elements which belongs to A but do not belong to B. In symbols,*

$$A - B := A \cap B^c.$$

Example 3.42. *Prove that*

$$A - B = B^c - A^c.$$

Proof. By the definition of the difference $A - B$, the commutative law, Theorem 3.33, and the definition of the difference $B^c - A^c$,

$$A - B = A \cap B^c = B^c \cap A = B^c \cap (A^c)^c = B^c - A^c.$$

□

Definition 3.43 (singleton and doubleton)**.** *(1) A class containing only a single element is called a "singleton".*

(2) A class containing two elements is called an "unordered pair" or a "doubleton".

Remark 3.44. Let a and b elements. Then the singleton

$$\{a\} = \{x : x = a\}$$

and the doubleton

$$\{a, b\} = \{x : x = a \ \vee \ x = b\}$$

exist due to the axiom of class construction. Therefore, if there exist elements a and b, then the classes singleton $\{a\}$ and doubleton $\{a, b\}$ exist. However, there exist a more fundamental question. " Is there an element?". We answer the question by adopting an appropriate axiom.

Axiom 3.45 (Axiom of the empty set)**.** *The empty class $\emptyset$ is a set.*

Theorem 3.46. *If $\{x, y\} = \{u, v\}$, then*

$$x = u \ \wedge \ y = v$$

or

$$x = v \ \wedge \ y = u.$$

Proof. Suppose that $\{x, y\} = \{u, v\}$. We complete the proof dividing it into the two cases that $x = y$ or $x \neq y$.

- (Case 1: $x = y$). Since $u \in \{u, v\}$ and $\{x, y\} = \{u, v\}$, $u \in \{x, y\}$ by the axiom of extent. Thus, by the definition of the doubleton, $u = x$ or $u = y$. Since $x = y$, we finally have $u = x = y$. Analogously, $v = x = y$. Therefore $u = v = x = y$.
- (Case 2: $x \neq y$). Since $x \in \{x, y\}$ and $\{x, y\} = \{u, v\}$, $x \in \{u, v\}$ by the axiom of extent. Thus, by the definition of the doubleton, $x = u$ or $x = v$. We consider the cases $x = u$ or $x = v$ separately.
  (i) ($x = u$). Since $y \in \{x, y\}$ and $\{x, y\} = \{u, v\}$, $y \in \{u, v\}$ by the axiom of extent. Thus, by the definition of the doubleton, $y = u$ or $y = v$. If $y = u$, then $y = u = x$ and it is contradiction to the fact that $x \neq y$. In other words, $y = u$ is impossible. Therefore we have $y = v$. In this case, we are done.
  (ii) ($x = v$). Since $y \in \{x, y\}$ and $\{x, y\} = \{u, v\}$, $y \in \{u, v\}$ by the axiom of extent. Thus, by the definition of the doubleton, $y = u$ or $y = v$. If $y = v$, then $y = v = x$ and it is contradiction to the fact that $x \neq y$. In other words, $y = v$ is impossible. Therefore we have $y = u$.

□

Definition 3.47. *Let a and b be elements. The ordered pair (a, b) is defined to be the class*

$$(a, b) := \{\{a\}, \{a, b\}\}.$$

For (a, b), a is called the "first component" of (a, b) and b is called the "second component" of (a, b).

Remark 3.48. (1) Observe that

$$(a, b) := \{\{a\}, \{a, b\}\} = \{x : x = \{a\} \text{ or } \{a, b\}\}.$$

The last equality above does not seem to be correct at this moment since there is no guarantee that $\{a\}$ and $\{a, b\}$ are elements. Therefore, the axiom of class construction does not imply the existence of the ordered pairs. We will introduce additional axioms to guarantee the existence.

(2) Note that
$$(b,a) := \{\{b\},\{b,a\}\}.$$
Therefore generally $(a,b) \neq (b,a)$.

Axiom 3.49 (Axiom of doubleton)**.** *If a and b are sets, then $\{a,b\}$ is a set.*

Remark 3.50. The easiest example of doubleton is $\{\emptyset,\emptyset\}$. Obviously, $\{\emptyset\} = \{\emptyset,\emptyset\}$. Thus by the axiom of doubleton and the axiom of the empty set, $\{\emptyset\}$ is a set. Moreover, applying the axiom of doubleton again, we can find another doubleton $\{\emptyset,\{\emptyset\}\}$.

Axiom 3.51 (Axiom of subset)**.** *Every subclass of a set is a set.*

Lemma 3.52. *Let a be an element. Then $\{a\}$ is a set.*

Proof. Recall that $\emptyset$ is a set by the axiom of the empty set. Moreover, $\{a,\emptyset\}$ is a set by the axiom of doubleton. Finally, since $\{a\} \subset \{a,\emptyset\}$, $\{a\}$ is a set by the axiom of subset. □

Theorem 3.53. *Let a and b be elements. Then (a,b) is a set*

Proof. Since a and b elements, the doubleton $\{a,b\}$ is a set by the axiom of doubleton. Moreover, $\{a\}$ is a set by Lemma 3.52. Therefore by the definition of the ordered pair (a,b) and the axiom of doubleton again, (a,b) is a set. □

Theorem 3.54. *If $(a,b) = (c,d)$, then $a = c$ and $b = d$.*

Proof. Suppose that $(a,b) = (c,d)$. Then by the definition of the ordered pairs,
$$\{\{a\},\{a,b\}\} = \{\{c\},\{c,d\}\}.$$
Then by Theorem 3.46,
$$\{a\} = \{c\} \ \wedge \ \{a,b\} = \{c,d\}$$
or
$$\{a\} = \{c,d\} \ \wedge \ \{a,b\} = \{c\}.$$
We consider these cases separately.

- (Case 1: $\{a\} = \{c\}$ and $\{a,b\} = \{c,d\}$). Since $\{a\} = \{c\}$, we have
$$a = c \tag{3.13}$$
by the definition of the singleton. From $\{a,b\} = \{c,d\}$, applying Theorem 3.46 again, we have
$$a = c \ \wedge \ b = d \quad \text{or} \quad a = d \ \wedge \ b = c.$$
In the first case, we have $a = c$ and $b = d$. Thus we are done. In the second case, by using (3.13), we have $d = a = c = b$. Thus the conclusion of the theorem holds.
- (Case 2: $\{a\} = \{c,d\}$ and $\{a,b\} = \{c\}$). Since $c \in \{c,d\} = \{a\}$, we have $c = a$. Similarly, since $d \in \{c,d\} = \{a\}$ and $b \in \{a,b\} = \{c\}$, we have $d = a$ and $b = c$. Therefore
$$b = c = a = d$$
and we are done.

□

Definition 3.55 (The Cartesian product)**.** *Let A and B classes. The "Cartesian product" of two classes A and B is the class of all ordered pairs (x, y) such that $x \in A$ and $y \in B$. In symbols,*

$$A \times B := \{(x, y) : x \in A \quad \textit{and} \quad y \in B\}.$$

For each $x \in A$ and $y \in B$, (x, y) is an element by Theorem 3.53. Therefore, the existence of the class $A \times B$ is guaranteed by the axiom of class construction.

Theorem 3.56. *Let A, B, and C be classes.*

(1)
$$A \times (B \cap C) = (A \times B) \cap (A \times C)$$
(2)
$$A \times (B \cup C) = (A \times B) \cup (A \times C)$$
(3)
$$(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$$

Proof. (1) By the definition of the Cartesian product, the associative law, Theorem 2.19(2), commutative law, and the definition of the intersection,

$$\begin{aligned}
(x, y) \in A \times (B \cap C) &\iff x \in A \quad \text{and} \quad y \in B \cap C \\
&\iff x \in A \quad \text{and} \quad (y \in B \quad \text{and} \quad y \in C) \\
&\iff x \in A \quad \text{and} \quad y \in B \quad \text{and} \quad y \in C \\
&\iff x \in A \quad \text{and} \quad x \in A \quad \text{and} \quad y \in B \quad \text{and} \quad y \in C \\
&\iff (x \in A \quad \text{and} \quad y \in B) \quad \text{and} \quad (x \in A \quad \text{and} \quad y \in C) \\
&\iff (x, y) \in (A \times B) \quad \text{and} \quad (x, y) \in (A \times C) \\
&\iff (x, y) \in (A \times B) \cap (A \times C).
\end{aligned}$$

(2) We leave the proof of (2) as an exercise.
(3) By the definition of the Cartesian product, the associative law, the commutative law, the definition of the intersection,

$$\begin{aligned}
(x, y) \in (A \times B) \cap (C \times D) &\iff (x, y) \in A \times B \quad \text{and} \quad (x, y) \in C \times D \\
&\iff x \in A \quad \text{and} \quad y \in B \text{ and} \quad x \in C \quad \text{and} \quad y \in D \\
&\iff (x \in A \text{ and} \quad x \in C) \quad \text{and} \quad (y \in B \quad \text{and} \quad y \in D) \\
&\iff x \in A \cap C \quad \text{and} \quad y \in B \cap D \\
&\iff (x, y) \in (A \cap C) \times (B \cap D).
\end{aligned}$$

□

Remark 3.57. It is often convenient to illustrate relations between products of classes by using a graphic device known as a "coordinate diagram". Here are examples of "coordinate diagram"

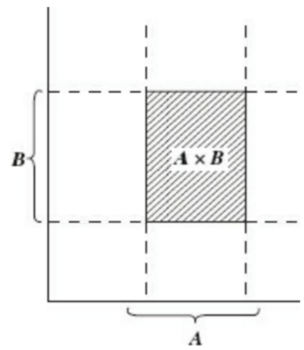

FIGURE 5. $A \times B$

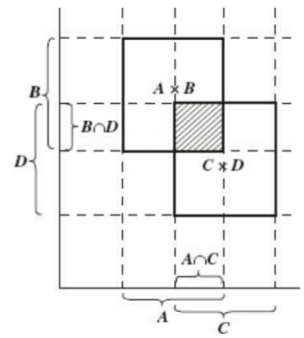

FIGURE 6. $(A \times B) \cap (C \times D)$ or $(A \cap C) \times (B \cap D)$

Coordinate diagrams are very useful to expect results. However, they do not give rigorous proofs.

Exercise 3.58. *Sketch coordinate diagrams for Theorem 3.56(1)-(2).*

Definition 3.59. *A class of ordered pairs is called a "graph". In other words, any subclass of $\mathcal{U} \times \mathcal{U}$ is called a "graph", where $\mathcal{U}$ is the universal class.*

Definition 3.60 (The inverse of a graph)**.** *Let G be a graph. Then G^{-1} is the graph defined by*

$$G^{-1} = \{(x, y) : (y, x) \in G\}.$$

Definition 3.61 (The composition of graphs)**.** *Let G and H be graphs. Then $G \circ H$ is the graph defined as follows:*

$$G \circ H := \{(x, y) : \exists z \ni (x, z) \in H \quad \text{and} \quad (z, y) \in G\}.$$

One can easily check that the classes G^{-1} and $G \circ H$ exist due to the axiom of class construction.

Theorem 3.62. *Let G, H, and J be graphs. Then*

(1) (associative property of the composition of graphs)

$$(G \circ H) \circ J = G \circ (H \circ J)$$

(2)

$$(G^{-1})^{-1} = G.$$

(3)

$$(G \circ H)^{-1} = H^{-1} \circ G^{-1}.$$

Proof. (1)

$$\begin{aligned}
(x, y) \in (G \circ H) \circ J &\iff \exists z \ni (x, z) \in J \quad \text{and} \quad (z, y) \in (G \circ H) \\
&\iff \exists z \ni (x, z) \in J \quad \text{and} \quad \exists w \ni (z, w) \in H \quad \text{and} \quad (w, y) \in G \\
&\iff (\exists z \text{ and } \exists w) \ni (x, z) \in J \quad \text{and} \quad (z, w) \in H \quad \text{and} \quad (w, y) \in G \\
&\iff \exists w \ni (\exists z \ni (x, z) \in J \quad \text{and} \quad (z, w) \in H) \quad \text{and} \quad (w, y) \in G \\
&\iff \exists w \ni (x, w) \in H \circ J \quad \text{and} \quad (w, y) \in G \\
&\iff (x, y) \in G \circ (H \circ J).
\end{aligned}$$

(2)
$$\begin{aligned}(x,y)\in(G^{-1})^{-1} &\iff (y,x)\in G^{-1}\\ &\iff (x,y)\in G.\end{aligned}$$

(3)
$$\begin{aligned}(x,y)\in(G\circ H)^{-1} &\iff (y,x)\in G\circ H\\ &\iff \exists z\ni(y,z)\in H\quad\text{and}\quad(z,x)\in G\\ &\iff \exists z\ni(z,y)\in H^{-1}\quad\text{and}\quad(x,z)\in G^{-1}\\ &\iff \exists z\ni(x,z)\in G^{-1}\quad\text{and}\quad(z,y)\in H^{-1}\\ &\iff (x,y)\in H^{-1}\circ G^{-1}.\end{aligned}$$

□

Definition 3.63. *Let G be a graph. By the "domain" of G, we mean the class*
$$dom\ G := \{x : \exists y\ni(x,y)\in G\}$$
and by the "range" of G, we mean the class
$$ran\ G := \{y : \exists x\ni(x,y)\in G\}.$$
In other words, the domain of G is the class of all "first components" of elements of G and the range of G is the class of all "second components" of elements of G.

Theorem 3.64. *Let G and H be graphs. Then*

(1)
$$dom\ G = ran\ G^{-1}$$
(2)
$$ran\ G = dom\ G^{-1}$$
(3)
$$dom\ (G\circ H)\subset dom\ H$$
(4)
$$ran\ (G\circ H)\subset ran\ G.$$

Proof. (1)
$$\begin{aligned}x\in dom\ G &\iff \exists y\ni(x,y)\in G\\ &\iff \exists y\ni(y,x)\in G^{-1}\\ &\iff x\in ran\ G^{-1}.\end{aligned}$$

(2) The proof of (2) is left as an exercise.

(3)
$$\begin{aligned}x\in dom\ (G\circ H) &\implies \exists y\ni(x,y)\in(G\circ H)\\ &\implies \exists y\ni(\exists z\ni(x,z)\in H\quad\text{and}\quad(z,y)\in G)\\ &\implies \exists y\text{ and }\exists z\ni(x,z)\in H\quad\text{and}\quad(z,y)\in G\\ &\implies \exists z\ni(x,z)\in H\\ &\implies x\in dom\ H.\end{aligned}$$

(4) The proof of (4) is left as an exercise.

□

Corollary 3.65. *Let G and H be graphs. Assume that $ran\ H \subset dom\ G$. Then*

$$dom\ G \circ H = dom\ H.$$

Proof. By Theorem 3.64(iii),

$$dom\ (G \circ H) \subset dom\ H.$$

Thus it is sufficient to show

$$dom\ H \subset dom\ (G \circ H).$$

Let $x \in dom\ H$. Then by the definition of the domain,

$$\exists y \ni (x, y) \in H. \tag{3.14}$$

By the definition of the range,

$$y \in ran\ H.$$

Moreover, by the assumption that $ran\ H \subset dom\ G$,

$$y \in dom\ G.$$

Thus by the definition of the domain of G,

$$\exists z \ni (y, z) \in G. \tag{3.15}$$

Finally, combining (3.14) and (3.15), we have

$$\begin{aligned} \exists y \ni (x, y) \in H \quad \text{and} \quad (y, z) \in G &\implies (x, z) \in G \circ H \\ &\implies x \in dom\ G \circ H. \end{aligned}$$

□

Definition 3.66 (Index class and indexed family of classes (Intuitive definition))**.** *A class I is called an "index class" if for every element $i \in I$, there exists a corresponding class A_i. For an index class I, $\{A_i : i \in I\}$ is called an "indexed family of classes".*

Definition 3.67 (Index class and indexed family of classes (Formal definition))**.** *An indexed family of classes denoted by $\{A_i\}_{i \in I}$ is a graph G whose domain is I. The domain I is called an "index class". For each $i \in I$, we define A_i by*

$$A_i := \{x : (i, x) \in G\}. \tag{3.16}$$

Example 3.68. *Consider $\{A_i\}_{i \in I}$ with $I = \{1, 2\}$, $A_1 = \{a, b\}$, and $A_2 = \{c, d\}$. Then, formally, $\{A_i\}_{i \in I}$ is the graph*

$$G = \{(1, a), (1, b), (2, c), (2, d)\}.$$

On the other hand, intuitively, the indexed family of class is given by

$$\{A_i : i \in I\} = \{A_1, A_2\} = \{\{a, b\}, \{c, d\}\}.$$

Remark 3.69. If $\{A_i\}_{i\in I}$ is given, then we can construct $\{A_i : i \in I\}$ due to (3.16). On the other hand, if $\{A_i : i \in I\}$ is given, then we can construct the graph

$$G = \{A_i\}_{i\in I} = \{(i,x) : i \in I \quad \text{and} \quad x \in A_i\}.$$

In this sense, we can identify $\{A_i : i \in I\}$ and $\{A_i\}_{i\in I}$. Therefore, we do not distinguish two concepts $\{A_i : i \in I\}$ and $\{A_i\}_{i\in I}$.

Definition 3.70. *Let $\{A_i\}_{i\in I}$ be an indexed family of classes.*

- *The "union of the classes A_i" is the class consisting of all the elements which belong to at least one class A_i of the family. In symbols,*
$$\bigcup_{i\in I} A_i := \{x : \exists j \in I \ni x \in A_j\}.$$
In other words, $x \in \bigcup_{i\in I} A_i$ iff $x \in A_j$ for some $j \in I$.
- *The "intersection of the classes A_i" is the class consisting of all the elements which belong to every class A_i of the family. In symbols,*
$$\bigcap_{i\in I} A_i := \{x : \forall i \in I,\ x \in A_i\}.$$
In other words, $x \in \bigcap_{i\in I} A_i$ iff $x \in A_i$ for all $i \in I$.

Theorem 3.71. *Let $\{A_i\}_{i\in I}$ be an indexed family of classes and B be an class. Then*

(1) If $A_i \subset B$ for all $i \in I$, then
$$\bigcup_{i\in I} A_i \subset B$$
(2) If $B \subset A_i$ for all $i \in I$, then
$$B \subset \bigcap_{i\in I} A_i.$$

Proof. (1) Suppose that $A_i \subset B$ for all $i \in I$ and let $x \in \bigcup_{i\in I} A_i$. Then there exists a $j \in I$ such that $x \in A_j$. Since $A_j \subset B$, we have
$$x \in B.$$
(2) Suppose that $B \subset A_i$ for all $i \in I$ and let $x \in B$. Then $x \in A_i$ for all $i \in I$. Finally, by the definition of the intersection of the classes A_i, we have
$$x \in \bigcap_{i\in I} A_i.$$
□

Theorem 3.72 (Generalized De Morgan's Laws)**.** *Let $\{A_i\}_{i\in I}$ be an indexed family of classes. Then*

(1)
$$\left(\bigcup_{i\in I} A_i\right)^c = \bigcap_{i\in I} A_i^c$$

(2)

$$\left(\bigcap_{i\in I} A_i\right)^c = \bigcup_{i\in I} A_i^c.$$

Proof. (1)

$$\begin{aligned} x \in \left(\bigcup_{i\in I} A_i\right)^c &\iff x \notin \bigcup_{i\in I} A_i \\ &\iff \text{there is no } j \in I \text{ such that } x \in A_j \\ &\iff \forall i \in I, x \notin A_i \\ &\iff \forall i \in I, x \in A_i^c \\ &\iff x \in \bigcap_{i\in I} A_i^c. \end{aligned}$$

(2)

$$\begin{aligned} x \in \left(\bigcap_{i\in I} A_i\right)^c &\iff x \notin \bigcap_{i\in I} A_i \\ &\iff \text{there exists a } j \in I \text{ such that } x \notin A_j \\ &\iff \exists j \in I \ni x \in A_j^c \\ &\iff x \in \bigcup_{i\in I} A_i^c. \end{aligned}$$

□

Remark 3.73. Recall the original De Morgan's law, i.e. for all classes A and B,

$$(A \cup B)^c = A^c \cap B^c$$

and

$$(A \cap B)^c = A^c \cup B^c.$$

Next let's consider three classes A_1, A_2, and A_3. Then applying the original De Morgan's law twice, we have

$$\begin{aligned} \left(\bigcup_{i=1}^{3} A_i\right)^c := (A_1 \cup A_2 \cup A_3)^c = ((A_1 \cup A_2) \cup A_3)^c &= (A_1 \cup A_2)^c \cap A_3^c \\ &= A_1^c \cap A_2^c \cap A_3^c \\ &=: \left(\bigcap_{i=1}^{3} A_i^c\right). \end{aligned}$$

Moreover, for any $n \in \mathbb{N}$, applying the original De Morgan's law $n-1$-times, we have

$$\left(\bigcup_{i=1}^{n} A_i\right)^c = \left(\bigcap_{i=1}^{n} A_i^c\right).$$

However, the original De Morgan's law does not guarantee that the generalized De Morgan's law since the number of elements of an index class can be infinite.

Exercise 3.74. *Let $\{A_n\}_{i\in\mathbb{N}}$ be an indexed family of classes. By using the original De Morgan's law, prove that*

$$\left(\bigcap_{i=1}^{\infty} A_i\right)^c := \left(\bigcap_{i\in\mathbb{N}} A_i\right)^c = \left(\bigcup_{i\in\mathbb{N}} A_i^c\right) =: \left(\bigcup_{i=1}^{\infty} A_i^c\right).$$

Theorem 3.75 (Generalized Distributive Laws)**.** *Let $\{A_i\}_{i\in I}$ and $\{B_j\}_{j\in J}$ be indexed families of classes. Then*

(1)

$$\left(\bigcup_{i\in I} A_i\right) \cap \left(\bigcup_{j\in J} B_j\right) = \bigcup_{(i,j)\in I\times J} (A_i \cap B_j)$$

(2)

$$\left(\bigcap_{i\in I} A_i\right) \cup \left(\bigcap_{j\in J} B_j\right) = \bigcap_{(i,j)\in I\times J} (A_i \cup B_j).$$

Proof. (1)

$$\begin{aligned}
x \in \left(\bigcup_{i\in I} A_i\right) \cap \left(\bigcup_{j\in J} B_j\right) &\iff x \in \bigcup_{i\in I} A_i \quad \text{and} \quad x \in \bigcup_{j\in J} B_j \\
&\iff \exists\ h \in I \ni x \in A_h \quad \text{and} \quad \exists\ k \in J \ni x \in B_k \\
&\iff \exists\ h \in I \text{ and } \exists k \in J \ni x \in A_h \text{ and } x \in B_k \\
&\iff \exists\ (h,k) \in I \times J \ni x \in A_h \cap B_k \\
&\iff x \in \bigcup_{(i,j)\in I\times J} (A_i \cap B_j).
\end{aligned}$$

(2)

$$\begin{aligned}
x \in \left(\bigcap_{i\in I} A_i\right) \cup \left(\bigcap_{j\in J} B_j\right) &\iff x \in \left(\bigcap_{i\in I} A_i\right) \text{ or } \left(\bigcap_{j\in J} B_j\right) \\
&\iff x \in A_i \ \forall i \in I \text{ or } x \in B_j \ \forall j \in J \\
&\iff x \in A_i \text{ or } x \in B_j \quad \forall (i,j) \in I \times J \\
&\iff x \in A_i \cup B_j \quad \forall (i,j) \in I \times J \\
&\iff \bigcap_{(i,j)\in I\times J} (A_i \cup B_j).
\end{aligned}$$

Exercise 3.76. *By using the contrapositive, prove that*

$$x \in A_i \text{ or } x \in B_j \quad \forall (i,j) \in I \times J \implies x \in A_i \ \forall i \in I \text{ or } x \in B_j \ \forall j \in J.$$

□

Theorem 3.77. *Let $\{G_i\}_{i\in I}$ be a family of graphs. Then*

(1)

$$dom\ \left(\bigcup_{i\in I} G_i\right) = \bigcup_{i\in I} (dom\ G_i)$$

(2)

$$ran\left(\bigcup_{i\in I} G_i\right) = \bigcup_{i\in I}(ran\ G_i).$$

Proof. (1)

$$\begin{aligned} x \in dom\left(\bigcup_{i\in I} G_i\right) &\iff \exists y \ni (x,y) \in \left(\bigcup_{i\in I} G_i\right) \\ &\iff \exists y \ni \ (\exists j \in I \ni (x,y) \in G_j) \\ &\iff \exists\ (y \text{ and } j \in I) \ni \ (x,y) \in G_j \\ &\iff \exists j \in I \ni \ (\exists y \ni (x,y) \in G_j) \\ &\iff \exists j \in I \ni \ (x \in dom\ G_j) \\ &\iff x \in \bigcup_{i\in I}(dom\ G_i). \end{aligned}$$

(2)

$$\begin{aligned} y \in ran\left(\bigcup_{i\in I} G_i\right) &\iff \exists x \ni (x,y) \in \bigcup_{i\in I} G_i \\ &\iff \exists x \ni (\exists j \in I \ni (x,y) \in G_j) \\ &\iff \exists j \in I \ni (\exists x \ni (x,y) \in G_j) \\ &\iff \exists j \in I \ni y \in ran\ G_j \\ &\iff y \in \bigcup_{i\in I}(ran\ G_i). \end{aligned}$$

□

Definition 3.78. *Let $\mathscr{A}$ be a class.*

(1) We define "the union of $\mathscr{A}$" to be the class of all elements which are contained for some class $A \in \mathscr{A}$. In symbols,

$$\bigcup_{A\in\mathscr{A}} A := \{x : \exists A \in \mathscr{A} \ni x \in A\}.$$

In other words, $x \in \bigcup_{A\in\mathscr{A}} A$ iff $x \in A$ for some class $A \in \mathscr{A}$.

(2) We define "the intersection of $\mathscr{A}$" to be the class of all elements which are contained for all classes $A \in \mathscr{A}$. In symbols,

$$\bigcap_{A\in\mathscr{A}} A := \{x : x \in A \quad \forall A \in \mathscr{A}\}.$$

In other words, $x \in \bigcap_{A\in\mathscr{A}} A$ iff for every class $A \in \mathscr{A}$, $x \in A$.

Notation 3.79. *Sometimes, we simply use $\bigcup\mathscr{A}$ and $\bigcap\mathscr{A}$ instead of $\bigcup_{A\in\mathscr{A}} A$ and $\bigcap_{A\in\mathscr{A}} A$, i.e.*

$$\bigcup\mathscr{A} := \bigcup_{A\in\mathscr{A}} A$$

and

$$\bigcap\mathscr{A} := \bigcap_{A\in\mathscr{A}} A.$$

Remark 3.80. Let $\mathscr{A}$ be a class. If there exists a set $A \in \mathscr{A}$, then $\bigcap \mathscr{A}$ is a set by the axiom of subset.

Axiom 3.81 (Axiom of union)**.** *If $\mathscr{A}$ is a set of sets, then $\bigcup \mathscr{A}$ is a set.*

Example 3.82. *Let a and b sets. Then we can consider the class $\{a, b\}$ and denote $c := \{a, b\}$. Then by the axiom the doubleton, $\{a, b\}$ is a set. Moreover, observe that*

$$\bigcup c = \bigcup_{A \in c} A = a \cup b.$$

Therefore by the axiom of union, $a \cup b$ is a set.

Definition 3.83 (The power set)**.** *Let A be a set. By the "power set" of A, we mean the class of all the subclasses of A. In symbols, the power set of A is denoted by $\mathcal{P}(A)$ and given by*

$$\mathcal{P}(A) = \{B : B \subset A\}.$$

Due to the axiom of subset, the power set is the class of all the sets B which satisfy $B \subset A$.

Remark 3.84. Consider two classes A and B. If $A \subset B$, then we may say that "the class B is larger than the class A". Next consider a set C. Generally, it is not true that $C \subset \mathcal{P}(C)$. However, by the definition of the power set of C, $\{a\} \in \mathcal{P}(C)$ for all $a \in C$. In this sense, we may say that the power set of C is larger than C. Therefore, there is no guarantee that $\mathcal{P}(C)$ is a set even though C is a set.

Axiom 3.85 (Axiom of power set)**.** *If A is a set, then the power set of A is a set.*

Theorem 3.86. *Let A and B sets. Then $A \times B$ is a set.*

Proof. Recall that

$$A \times B = \{(x, y) : x \in A \quad \text{and} \quad y \in B\}$$

and

$$(x, y) = \{\{x\}, \{x, y\}\}.$$

We claim

$$A \times B \subset \mathcal{P}\left[\mathcal{P}(A \cup B)\right]. \tag{3.17}$$

Let $(x, y) \in A \times B$. Since $\{x\} \subset A \subset A \cup B$ and $\{x, y\} \subset A \cup B$, by the definition of the power set of $A \cup B$, $\{x\} \in \mathcal{P}(A \cup B)$ and $\{x, y\} \in \mathcal{P}(A \cup B)$. Thus

$$\{\{x\}\} \subset \mathcal{P}(A \cup B), \ \{\{x, y\}\} \subset \mathcal{P}(A \cup B), \ \text{and} \ \{\{x\}, \{x, y\}\} \subset \mathcal{P}(A \cup B).$$

Moreover by the definition of the power set, we have

$$\{\{x\}, \{x, y\}\} \in \mathcal{P}\left[\mathcal{P}(A \cup B)\right]. \tag{3.18}$$

Therefore, the claim is proved. Note that by the axiom of union and the axiom of power set, $\mathcal{P}\left[\mathcal{P}(A \cup B)\right]$ is a set. Finally by the axiom of subset and (3.17), $A \times B$ is a set □

Corollary 3.87. *Let A and B sets. Then $\bigcup (A \times B)$ is a set.*

Proof. By Theorem 3.86, $A \times B$ is a set. Thus by the axiom of union, $\bigcup (A \times B)$ is a set. □

Corollary 3.88. *Let A and B sets. Then for all $a_1, a_2 \in A$ and $b_1, b_2 \in B$,*

$$\{\{a_1\}, \{a_2\}, \{a_1, b_1\}, \{a_2, b_2\}\}$$

is a set.

Proof. Since $(a_1, b_1) \in A \times B$ and $(a_2, b_2) \in A \times B$,

$$(a_1, b_1) \cup (a_2, b_2) \subset \bigcup (A \times B).$$

Recall that $\bigcup(A \times B)$ is a set by Corollary 3.87. Thus by the axiom of subset, $(a_1, b_1) \cup (a_2, b_2)$ is a set. Moreover, by the definition of the ordered pair,

$$(a_1, b_1) \cup (a_2, b_2) = \{\{a_1\}, \{a_1, b_1\}\} \cup \{\{a_2\}, \{a_2, b_2\}\}.$$

Finally, by the definition of union and the axiom of extent, we have

$$\begin{aligned} \{\{a_1\}, \{a_2\}, \{a_1, b_1\}, \{a_2, b_2\}\} &= \{\{a_1\}, \{a_1, b_1\}\} \cup \{\{a_2\}, \{a_2, b_2\}\} \\ &= (a_1, b_1) \cup (a_2, b_2) \end{aligned}$$

is a set.

□

Exercise 3.89. *Let G be a graph. Assume that G is a set. Prove*

(1) dom G is a set

(2) Ran G is a set

(Hint: Consider $\bigcup(\bigcup G)$).

4. FUNCTIONS

Definition 4.1 (function (intuitive definition))**.** *Let A and B be classes. Assume that for each element x of A, there is an associated element of B, which is denoted by $f(x)$. Then f is said to be a "function from A to B (or a* **mapping** *of A to B)".*

Let f be a function from A to B. Then for each $x \in A$, there exists an associate element $f(x)$. Consider the graph

$$G = \{(x, f(x)) : x \in A\} \subset \{(x, y) : x \in A \text{ and } y \in B\} = A \times B.$$

Obviously, this graph G has all information about the function f from A to B. In other words, we can understand functions as graphs.

Definition 4.2 (function (formal definition))**.** *Let A and B be classes, and let f be a subclass of $A \times B$. We say that a triple $\langle f, A, B\rangle$ is a "function from A to B" if the following two properties hold:*

(F1).

$$\forall x \in A,\ \exists y \in B \text{ such that } (x, y) \in f$$

(F2). *If $(x, y_1) \in f$ and $(x, y_2) \in f$, then $y_1 = y_2$.*

Notation 4.3.
- *It is customary to use $f : A \to B$ instead of $\langle f, A, B\rangle$. We simply say $f : A \to B$ is a function instead of calling it a function from A to B. Moreover, many times we call the graph f a function instead of whole triple $\langle f, A, B\rangle$.*

- *For two functions* $f : A \to B$ *and* $g : C \to D$*, we write*
$$f : A \to B = g : C \to D \tag{4.1}$$
if $A = C$*,* $B = D$*, and* $f = g$*. We say that two functions* $f : A \to B$ *and* $g : C \to D$ *are equal if* (4.1) *holds.*

Remark 4.4. The property (F2) can be extended for any graph G. For a graph G, we say that G satisfies $(F2)$ if $(x, y_1) \in G$ and $(x, y_2) \in G$, then $y_1 = y_2$.

Lemma 4.5. *Let* $f : A \to B$ *be a function. Then*
(1) $dom\ f = A$
(2) $ran\ f \subset B$.

Proof. (1) Let $x \in dom\ f$. Then by the definition of the domain, there exists y such that $(x, y) \in f$. Since $f \subset A \times B$, we have $(x, y) \in f \subset A \times B$. Thus $x \in A$. Next, let $x \in A$. Then by (F1), there exists a $y \in B$ such that $(x, y) \in f$. Therefore by the definition of the domain of f,
$$x \in dom\ f.$$
(2) Let $y \in ran\ f$. Then by the definition of the range, there exists a x such that $(x, y) \in f$. Since $f \subset A \times B$, $(x, y) \in f \subset A \times B$. Therefore, by the definition of Cartesian product, $y \in B$.

□

Notation 4.6. *Let* $f : A \to B$ *be a function. Then* A *is the domain of the graph* f *by Lemma 4.5. Moreover, we say that* A *is the domain of the function* f*. On the other hand,* B *is called the "codomain" of the function* f *or the "target space" of the function* f*. We simply say that* A *is the domain and* B *is the codomain (or target space) if the given function is clear.*

Definition 4.7. *Let* $f : A \to B$ *be a function. If* $(x, y) \in f$*, then we say that*
(1) y *is the image of* x *(with respect to* f*)*
(2) x *is the pre-image of* y *(with respect to* f*)*
(3) f *maps* x *onto* y *and it is symbolized by* $x \overset{f}{\mapsto} y$ *or simply* $x \mapsto y$*.*

According to these definitions,
- *(F1) states that every element* $x \in A$ *has an image* $y \in B$
- *(F2) states that if* $x \in A$*, then the image of* x *is unique*
- *(F1) and (F2) state that every element* $x \in A$ *has a unique image* $y \in B$*.*

Theorem 4.8. *Let* A *and* B *be classes, and let* f *be a graph. Then* $f : A \to B$ *is a function if and only if the following three properties hold:*
(1) f *satisfies (F2)*
(2) $dom\ f = A$
(3) $ran\ f \subset B$.

Proof. The only if part is obvious due to Lemma 4.5 and the definition of the function. Thus we only prove the if part. Assume that the graph f and classes A

and B satisfy (1), (2), and (3). First we prove that $f \subset A \times B$. Let $(x, y) \in f$. Then

$$x \in dom\ f \quad \text{and} \quad y \in ran\ f.$$

Thus by (2) and (3),

$$x \in A \quad \text{and} \quad y \in B.$$

Thus by the definition of the Cartesian product,

$$(x, y) \in A \times B.$$

Next we show that (F1) holds. Let $x \in A$. Then by (2), $x \in dom\ f$. Thus by the definition of the domain of f, there exists a y such that $(x, y) \in f$. It only remains to show that $y \in B$. Observe that $y \in ran\ f$. Therefore by (3), $y \in B$. Finally, it is obvious that (F2) holds due to (1). □

Corollary 4.9. *Let $f : A \to B$ be a function and C be a class such that $ran\ f \subset C$. Then $f : A \to C$ is a function.*

Proof. Since $f : A \to B$ is a function, due to Theorem 4.8, (F2) holds, $dom\ f = A$, and $ran\ f \subset B$. Moreover, recall the assumption that $ran\ f \subset C$. Thus, (F2) holds, $dom\ f = A$, and $ran\ f \subset C$. In other words, (1), (2), and (3) of Theorem 4.8 hold for the classes A and C, and the graph f. Therefore, $f : A \to C$ is a function. □

Notation 4.10. *Let $f : A \to B$ be a function and $x \in A$. It is customary to use the symbol $f(x)$ to designate the (unique) image of x. Then,*

$$y = f(x) \quad \iff \quad (x, y) \in f. \tag{4.2}$$

By using this notation, (F1) and (F2) can be restated as follows:

(F1) $\forall x \in A, \exists y \in B$ *such that* $y = f(x)$.
(F2) *If* $y_1 = f(x)$ *and* $y_2 = f(x)$, *then* $y_1 = y_2$.

Moreover, (F2) is equivalent to the sentence that

(F2') *if* $x_1, x_2 \in dom\ f$ *and* $x_1 = x_2$, *then* $f(x_1) = f(x_2)$.

Sometimes for a graph G and $x \in dom\ G$, we use the notation (definition)

$$G(x) = y \quad \iff \quad y = G(x) \quad \iff \quad (x, y) \in G. \tag{4.3}$$

Exercise 4.11. *Let $f : A \to B$ be a function. Prove that (F2) iff (F2').*

Exercise 4.12. *Let G be a graph. Prove or disprove the followings:*

(1) (F2) only if (F2')
(2) (F2) if (F2').

Theorem 4.13. *Let $f : A \to B$ and $g : A \to B$ be functions. Then $f = g$ if and only if $f(x) = g(x)$ for all $x \in A$.*

Proof. We prove the only if part first. Assume that $f = g$. Then by the axiom of extent,

$$(x, y) \in f \iff (x, y) \in g. \tag{4.4}$$

Let $x \in A$. Then there exist $y_f, y_g \in B$ such that $y_f = f(x)$ and $y_g = g(x)$. Moreover by (4.2) and (4.4),

$$y_f = g(x).$$

Therefore, by (F2)

$$g(x) = y_g = y_f = f(x).$$

Next we prove the if part. Assume that $f(x) = g(x)$ for all $x \in A$. Then by (4.2),

$$(x, y) \in f \iff y = f(x) \iff y = g(x) \iff (x, y) \in g.$$

Therefore, $f = g$ by the axiom of the extent. □

Definition 4.14. *(1) A function $f : A \to B$ is said to be "injective" if it has the following property:*

(INJ) *if $(x_1, y) \in f$ and $(x_2, y) \in f$, then $x_1 = x_2$.*

(2) A function $f : A \to B$ is said to be "surjective" if it has the following property:

(SURJ) *for all $y \in B$, there exists a $x \in A$ such that $y = f(x)$.*

(3) A function $f : A \to B$ is said to be "bijective" if it is both injective and surjective.

(4) If there exists a bijective function $f : A \to B$, then we say that two classes A and B are in "ono-to-one correspondence".

Remark 4.15. (1) (INJ) states that y has no more than one pre-image.
(2) (INJ) is equivalent to the sentence that if $f(x_1) = f(x_2)$, then $x_1 = x_2$.
(3) (SURJ) states that every element of B is the image of at least one element of A. It obviously implies that $B \subset ran\ f$. On the other hand, $ran\ f \subset B$ by Theorem 4.8. Therefore, (SURJ) holds iff $B = ran\ f$.
(4) Combining (1) and (3), we can say that if f is bijective, then every element of B has exactly one pre-image in A.
(5) We can extend the definition of injective and surjective properties for a general graph. Indeed, we say that a graph G is injective if

$$(x_1, y) \in f \text{ and } (x_2, y) \in f \implies x_1 = x_2.$$

A graph G is said to be surjective to a class B if for all $y \in B$, there exists a $x \in dom\ G$ such that $y = G(x)$. In particular, if $G : A \to B$ is a function, we just say that the graph G is surjective if the function $G : A \to B$ is surjective.

Exercise 4.16. *Prove that any graph G is surjective to ran G.*

Example 4.17 (Identity function)**.** *Let A be a class. By the "identity function on A", we mean the function $I_A : A \to A$ given by*

$$I_A(x) = x, \qquad \forall x \in A.$$

In other words,

$$I_A = \{(x, x) : x \in A\}.$$

First we show that $I_A : A \to A$ is injective. Let $I_A(x_1) = I_A(x_2)$. Then

$$x_1 = I_A(x_1) = I_A(x_2) = x_2.$$

Thus it is injective. Next we prove that $I_A : A \to A$ is surjective. Let $x \in A$. Then $I_A(x) = x$. Therefore, it is surjective. In total, the identity function $I_A : A \to A$ is bijective.

Example 4.18 (Constant function)**.** *Let A and B be classes, and let b be an element of B. By the "constant function" K_b, we mean the function $K_b : A \to B$ such that*

$$K_b(x) = b, \qquad \forall x \in A.$$

In other words, $K_b = \{(x, b) : x \in A\}$. In general, the constant functions are not injective nor surjective. Indeed, if A has more than one element, then K_b is not injective. If B has more than one element, then K_b is not surjective.

Example 4.19 (Inclusion Function)**.** *Let A be a class and B be a subclass of A. By the "inclusion function" of B in A, we mean the function $E_B : B \to A$ given by*

$$E_B(x) = x, \qquad \forall x \in B.$$

In other words, $E_B = \{(x, x) : x \in B\}$. It is easy to check that E_B is injective. Generally, it is not surjective. However, if $B = A$, then the inclusion function coincides with the identity function I_A and thus bijective.

Example 4.20 (Characteristic function (Indicator function))**.** *Let 2 be a class consisting of two elements and put $2 = \{0, 1\}$. If A is a class and B is a subclass of A, then the "characteristic function (or indicator function)" of B in A is the function $C_B : A \to 2$ defined as*

$$C_B(x) = \begin{cases} 1 & \textit{if} \quad x \in B \\ 0 & \textit{if} \quad x \notin B \textit{ and } x \in A. \end{cases}$$

In other words,

$$C_B = \{(x, 1) : x \in B\} \cup \{(x, 0) : x \notin B \textit{ and } x \in A\}.$$

It is obvious that the C_B maps every element of B onto 1 and every element of $A - B$ onto 0.

Example 4.21 (Restriction of a function)**.** *Let $f : A \to B$ be a function and C be a subclass of A. By the "restriction of f to C", we mean the function $f_{[C]} : C \to B$ such that*

$$f_{[C]}(x) = f(x) \qquad \forall x \in C. \tag{4.5}$$

In other words,

$$f_{[C]} = \{(x, y) : (x, y) \in f \textit{ and } x \in C\}.$$

Example 4.22 (Extension of a function)**.** *Let $f : A \to B$ be a function and C be a class such that $A \subset C$. By an "extension of f to C", we mean a function $f^{[C]} : C \to B$ such that*

$$f^{[C]}(x) = f(x) \qquad \forall x \in A. \tag{4.6}$$

In particular, the function f is the restriction of $f^{[C]}$ to A. Generally, the extension of f is not unique.

Exercise 4.23. *Let $f : A \to B$ be a function and C be a class such that $A \subsetneq C$. Prove*

(1) there exists an extension of $f : A \to B$ to C

(2) *the extension of* $f : A \to B$ *to* C *is not unique, i.e. at least there exist two different extensions of* $f : A \to B$ *to* C*.*

Lemma 4.24. *Let* $f_1 : B \to A$ *and* $f_2 : C \to A$ *be functions, where* $B \cap C = \emptyset$*. Define* $f = f_1 \cup f_2$*. Then*

(1) $(x, y) \in f$ *and* $x \in B$ *iff* $(x, y) \in f_1$
(2) $(x, y) \in f$ *and* $x \in C$ *iff* $(x, y) \in f_2$
(3) *dom* $f = B \cup C$ *and ran* $f = (ran\ f_1) \cup (ran\ f_2) \subset A$
(4) *If* $x \in B$ *then* $f(x) = f_1(x)$ *and if* $x \in C$ *then* $f(x) = f_2(x)$
(5) f *satisfies (F2).*

Proof. (1) First we prove the if part. Let $(x, y) \in f_1$. Then obviously $(x, y) \in f$ since $f_1 \subset f$. By the definition of the domain, $x \in dom\ f_1$. Moreover, $dom\ f_1 = B$ since $f_1 : B \to A$ is a function (cf. Theorem 4.8). Therefore, $x \in B$. Conversely, we assume $(x, y) \in f$ and $x \in B$. Since $(x, y) \in f$ and $f = f_1 \cup f_2$, $(x, y) \in f_1$ or $(x, y) \in f_2$. It is sufficient to show that $(x, y) \in f_2$ is impossible. Suppose $(x, y) \in f_2$. Then $x \in dom\ f_2 = C$. Since $C \cap B = \emptyset$, $x \notin B$. But it is contradiction to the assumption that $x \in B$. Thus $(x, y) \notin f_2$.

(2) The proof of (2) is analogous to that of (1). We skip the detail.

(3) By Theorem 3.77 and Theorem 4.8,

$$dom\ f = dom\ (f_1 \cup f_2) = (dom\ f_1) \cup (dom\ f_2) = B \cup C.$$

Similarly,

$$ran\ f = ran\ (f_1 \cup f_2) = (ran\ f_1) \cup (ran\ f_2) \subset A \cup A = A.$$

(4) Let $x \in B$. Then since $f_1 : B \to A$ is a function, $(x, f_1(x)) \in f_1$. Moreover, since $f_1 \subset f$, $(x, f_1(x)) \in f$. Therefore, recalling (4.3), we have $f(x) = f_1(x)$. Similarly, if $x \in C$, then $f(x) = f_2(x)$.

(5) Let $(x, y_1) \in f$ and $(x, y_2) \in f$. Then since $f = f_1 \cup f_2$, there exist four cases : (Case 1) $(x, y_1) \in f_1$ and $(x, y_2) \in f_2$, (Case 2) $(x, y_1) \in f_2$ and $(x, y_2) \in f_1$, (Case 3) $(x, y_1) \in f_1$ and $(x, y_2) \in f_1$, (Case 4) $(x, y_1) \in f_2$ and $(x, y_2) \in f_2$. First we prove that (Case 1) and (Case 2) cannot happen. Since f_1 is a function from B to A, $dom\ f_1 = B$. Similarly, $dom\ f_2 = C$. Moreover, the assumption $B \cap C = \emptyset$ implies that $x \in B$ and $x \in C$ are impossible. Therefore, (Case 1) and (Case 2) are impossible. It only remains to consider (Case 3) and (Case 4). Recall that (Case 3: $(x, y_1) \in f_1$ and $(x, y_2) \in f_1$. Since $f_1 : B \to A$ is a function, $y_1 = y_2$. Similarly, for (Case 4) $(x, y_1) \in f_2$ and $(x, y_2) \in f_2$, we have $y_1 = y_2$ since $f_2 : C \to A$ is a function.

$\square$

Exercise 4.25. *Under the setting Lemma 4.24, prove that (F2') holds for* f *without using Lemma 4.24(5).*

Theorem 4.26. *Let* $f_1 : B \to A$ *and* $f_2 : C \to A$ *be functions, where* $B \cap C = \emptyset$*. Define* $f = f_1 \cup f_2$*. Then*

(1) $f : B \cup C \to A$ *is a function*
(2) $f_1 = f_{[B]}$ *and* $f_2 = f_{[C]}$
(3) $f : B \cup C \to A$ *is an extension of* f_1 *to* $B \cup C$

(4) *$f : B \cup C \to A$ is an extension of f_2 to $B \cup C$.*

Proof. (1) Due to Lemma 4.24(3), (5), and Theorem 4.8, $f : B \cup C \to A$ is a function.

(2) By (1), $f : B \cup C \to A$ is a function. Thus due to Lemma 4.24(4),

$$f_1(x) = f(x) \qquad \forall x \in B \tag{4.7}$$

and

$$f_2(x) = f(x) \qquad \forall x \in C.$$

Therefore, by (4.5), we have $f_1 = f_{[B]}$ and $f_2 = f_{[C]}$

(3) By (4.7) and (4.6), $f : B \cup C \to A$ is an extension of f_1 to $B \cup C$.

(4) The proof of (4) is very analogous to that of (3). We skip the detail. □

Theorem 4.27. *Let $f : A \to B$ and $g : B \to C$ be functions. Then $g \circ f : A \to C$ is a function.*

Proof. Due to Theorem 4.8, it is sufficient to show that $g \circ f$ satisfies (F2), *dom* $g \circ f = A$, and *ran* $g \circ f \subset C$. First we prove *ran* $g \circ f \subset C$. By Theorem 3.64(4) and Theorem 4.8,

$$ran\ g \circ f \subset ran\ g \subset C.$$

Next we show *dom* $g \circ f = A$. By Theorem 4.8, *ran* $f \subset B =$ *dom* g. Thus by Corollary 3.65 and Theorem 4.8,

$$dom\ g \circ f = dom\ f = A.$$

It only remains to show that $g \circ f$ satisfies (F2). Let $(x, y_1) \in g \circ f$ and $(x, y_2) \in g \circ f$. Then by the definition of the composition, there exist z_1 and z_2 such that

$$(x, z_1) \in f \quad \wedge \quad (z_1, y_1) \in g$$

and

$$(x, z_2) \in f \quad \wedge \quad (z_2, y_2) \in g.$$

Since $f : A \to B$ is a function, $(x, z_1) \in f$ and $(x, z_2) \in f$ implies $z_1 = z_2$. Thus since $g : B \to C$ is a function, $(z_1, y_1) \in g$ and $(z_2, y_2) = (z_1, y_2) \in g$ implies $y_1 = y_2$. Therefore $g \circ f$ satisfies (F2). □

Exercise 4.28. *Let $f : A \to B$ and $g : B \to C$ be functions. Prove that*

$$x \xmapsto{g \circ f} y \quad \iff \quad \exists z \ni x \xmapsto{f} z \text{ and } z \xmapsto{g} y.$$

Exercise 4.29. *Let $f : A \to B$ and $g : B \to C$ be functions. Prove that*

$$g \circ f(x) = g(f(x)) \qquad \forall x \in A.$$

Definition 4.30. *A function $f : A \to B$ is said to be "invertible" if $f^{-1} : B \to A$ is a function.*

Lemma 4.31. *Let $f : A \to B$ be a invertible function. Then*

$$y = f(x) \qquad \iff \qquad x = f^{-1}(y).$$

Proof. By (4.2) and the definition of the inverse of the graph f,

$$y = f(x) \iff (x, y) \in f \iff (y, x) \in f^{-1} \iff x = f^{-1}(y).$$

□

Exercise 4.32. *Let $f : A \to B$ be a function. Prove or disprove that*

$$y = f(x) \qquad \iff \qquad x = f^{-1}(y).$$

Lemma 4.33. *Let $f : A \to B$ be a bijective function. Then $f^{-1} : B \to A$ is a function.*

Proof. Due to Theorem 4.8, it is sufficient to show that (1) $dom\ f^{-1} = B$, (2) $ran\ f^{-1} \subset A$, and (3) f^{-1} satisfies (F2).

(1) Let $y \in dom\ f^{-1}$. Then by the definition of the domain, there exists a x such that $(y, x) \in f^{-1}$. Moreover, by the definition of the inverse of graph, $(x, y) \in f$. Since $f : A \to B$ is a function, $y \in ran\ f \subset B$ by the definition of the range and Theorem 4.8. Conversely, let $y \in B$. Then since $f : A \to B$ is surjective, there exists a $x \in A$ such that $(x, y) \in f$. Moreover, by the definition of the inverse of graph, $(y, x) \in f^{-1}$. Therefore, by the definition of the domain, $y \in dom\ f^{-1}$.

(2) Let $x \in ran\ f^{-1}$. Then by the definition of the range, there exists a y such that $(y, x) \in f^{-1}$. By the definition of the inverse, $(x, y) \in f$. Thus $x \in dom\ f$ by the definition of the domain. Moreover, since $f : A \to B$ is a function, $x \in dom\ f = A$ by Theorem 4.8.

(3) Assume that $(x, y_1) \in f^{-1}$ and $(x, y_2) \in f^{-1}$. Then $(y_1, x) \in f$ and $(y_2, x) \in f$. Since $f : A \to B$ is injective, $y_1 = y_2$. Therefore (F2) holds for f^{-1}.

□

Theorem 4.34. *Let $f : A \to B$ be a bijective function. Then $f^{-1} : B \to A$ is a bijective function.*

Proof. Due to Lemma 4.33, $f^{-1} : B \to A$ is a function. Thus it is sufficient to show that the function $f^{-1} : B \to A$ is bijective. Let $x \in A$. Then since $f : A \to B$ is a function, there exists a $y \in B$ such that $(x, y) \in f$. Thus $(y, x) \in f^{-1}$ and it implies that $f^{-1}(y) = x$. Therefore, the function $f^{-1} : B \to A$ is surjective. To prove that the function $f^{-1} : B \to A$ is injective, let $(x_1, y) \in f^{-1}$ and $(x_2, y) \in f^{-1}$. Then $(y, x_1) \in f$ and $(y, x_2) \in f$ by the definition of the inverse. Since $f : A \to B$ is a function, $x_1 = x_2$ by (F2). □

Theorem 4.35. *Let $f : A \to B$ is a invertible function. Then $f : A \to B$ is bijective.*

Proof. Since $f : A \to B$ is invertible, $f^{-1} : B \to A$ is a function. Thus by Theorem 4.8, $dom\ f^{-1} = B$. By Theorem 3.64(2), $dom\ f^{-1} = B$ implies $ran\ f = dom\ f^{-1} = B$. Thus $f : A \to B$ is surjective. To prove that $f : A \to B$ is injective, assume that $(x_1, y) \in f$ and $(x_2, y) \in f$. Then $(y, x_1) \in f^{-1}$ and $(y, x_2) \in f^{-1}$. Since $f^{-1} : B \to A$ is a function, we have $x_1 = x_2$ by (F2). □

Corollary 4.36. *Let $f : A \to B$ be a function. Then $f : A \to B$ is invertible if and only if f is bijective. Moreover, if $f : A \to B$ is invertible, then $f^{-1} : B \to A$ is a bijective function.*

Theorem 4.37. *Let $f : A \to B$ be an invertible function. Then*
(1) $f^{-1} \circ f = I_A$
(2) $f \circ f^{-1} = I_B$.

Proof. (1) Let $(x, y) \in f^{-1} \circ f$. Then there exists a z such that $(x, z) \in f$ and $(z, y) \in f^{-1}$. Moreover, by the definition of the inverse of graph, $(y, z) \in f$. Thus we have $(x, z) \in f$ and $(y, z) \in f$. Since f is injective due to Corollary 4.36, we obtain

$$x = y.$$

Since $f : A \to B$ is a function, $x \in dom\ f = A$. Therefore, by the definition of I_A, $(x, y) = (x, x) \in I_A$. Conversely, let $(x, y) \in I_A$. Then by the definition of I_A, $x = y$ and $x \in A$. It is sufficient to show that $(x, x) \in f^{-1} \circ f$. Since $f : A \to B$ is a function, there exists a z such that $f(x) = z$, i.e. $(x, z) \in f$. By Lemma 4.31, $x = f^{-1}(z)$. Thus $(z, x) \in f^{-1}$. Therefore, by the definition of the composition of graphs, $(x, x) \in f^{-1} \circ f$.

(2) Since the proof of (2) is very similar to (1), we skip the detail. □

Exercise 4.38. *Prove Theorem 4.37 by using Theorem 4.13.*

Theorem 4.39. *Let $f : A \to B$ and $g : B \to A$ be functions. Assume that $g \circ f = I_A$ and $f \circ g = I_B$. Then $f : A \to B$ is bijective.*

Proof. First we prove that f is injective. Let $(x_1, y) \in f$ and $(x_2, y) \in f$. Then $y \in ran\ f \subset B$. Since $g : B \to A$ is a function, there exists $g(y) \in A$ such that

$$(y, g(y)) \in g.$$

By the definition of the composition $g \circ f$,

$$(x_1, g(y)) \in g \circ f \quad \text{and} \quad (x_2, g(y)) \in g \circ f.$$

Thus by the assumption that $g \circ f = I_A$, we have

$$(x_1, g(y)) \in I_A \quad \text{and} \quad (x_2, g(y)) \in I_A.$$

Moreover, by the definition of I_A,

$$(x_1, g(y)) = (x_3, x_3) \quad \text{and} \quad (x_2, g(y)) = (x_4, x_4)$$

for some $x_3, x_4 \in A$. Thus applying Theorem 3.54, we obtain

$$x_1 = x_3 = g(y) = x_4 = x_2.$$

Next we prove that $f : A \to B$ is surjective. Let $y \in B$. Then by the definition of I_B, $(y, y) \in I_B$. Moreover by the assumption that $f \circ g = I_B$, we have $(y, y) \in f \circ g$. Thus by the definition of the composition of graphs, there exists a z such that $(y, z) \in g$ and $(z, y) \in f$. Therefore, $y = f(z)$. □

Exercise 4.40. *Let $f : A \to B$ and $g : B \to A$ be functions. Assume that $g \circ f = I_A$. Then f is injective.*

Exercise 4.41. *Let $f : A \to B$ and $g : B \to A$ be functions. Assume that $f \circ g = I_B$. Then f is surjective.*

Corollary 4.42. *Let $f : A \to B$ and $g : B \to A$ be functions. If $g \circ f = I_A$ and $f \circ g = I_B$, then f is invertible and $f^{-1} = g$.*

Proof. Due to Theorem 4.39 and Corollary 4.36, it only remains to show that $f^{-1} = g$. Moreover, by Theorem 4.13, it is sufficient to show that

$$f^{-1}(y) = g(y) \qquad \forall y \in B.$$

Let $y \in B$ and recall that $f^{-1} : B \to A$ is a (bijective) function by Corollary 4.36. Then since both $f^{-1} : B \to A$ and $g : B \to A$ are functions, there exists $x_1 \in A$ and $x_2 \in A$ such that $x_1 = f^{-1}(y)$ and $x_2 = g(y)$. Due to Lemma 4.31, $y = f(x_1)$. By Exercise 4.29, the assumption that $g \circ f = I_A$ with Theorem 4.13, and the definition of the identity function on A,

$$g(y) = g \circ f(x_1) = I_A(x_1) = x_1.$$

Therefore, since $x_2 = g(y)$ and $g : B \to A$ is a function. $x_1 = x_2$, i.e. $f^{-1}(y) = g(y)$. □

Corollary 4.43. *Let $f : A \to B$ be a function. Then $f : A \to B$ is invertible if and only if there exists a function $g : B \to A$ such that $g \circ f = I_A$ and $f \circ g = I_B$. Moreover, $g = f^{-1}$.*

Exercise 4.44. *Prove Corollary 4.43.*

Exercise 4.45. *Let $f : A \to B$ be an invertible function. Then $f^{-1} : B \to A$ is a function. Prove that $f^{-1} : B \to A$ is invertible.*

Definition 4.46. *Let $f : A \to B$ be a function.*

1. *We call a function $g : B \to A$ a left inverse of $f : A \to B$ if $g \circ f = I_A$.*
2. *We call a function $g : B \to A$ a right inverse of $f : A \to B$ if $f \circ g = I_B$.*

Theorem 4.47. *Let $f : A \to B$ be a function and assume that $A \neq \emptyset$. Then $f : A \to B$ is injective if and only if there exists a left inverse of $f : A \to B$, i.e. there exists a function $g : B \to A$ such that $g \circ f = I_A$.*

Proof. Since the if part comes from Exercise 4.40, we only prove the only if part. Assume that $f : A \to B$ is injective and let $C = ran\ f$. Then by Corollary 4.9, $f : A \to C$ is a function. It is injective by the assumption and moreover it is surjective since $C = ran\ f$. Thus $f : A \to C$ is bijective and $f^{-1} : C \to A$ is a bijective function by Corollary 4.36. Moreover, by Corollary 4.43 and Exercise 4.29,

$$f^{-1}(f(x)) = I_A(x) \qquad \forall x \in A. \tag{4.8}$$

Since A is a nonempty class, there exists a $a \in A$. Consider the constant function $K_a : B - C \to A$ (cf. Example 4.18). In particular, recall that K_a is the graph given by $K_a = \{(x, a) : x \in B - C\}$. Define the graph $g = f^{-1} \cup K_a$. Then by Theorem 4.26(1), $g : B \to A$ is a function. We claim that g is a left inverse of $f : A \to B$. To prove this, it is sufficient to show that

$$g \circ f(x) = I_A(x) \qquad \forall x \in A$$

due to Theorem 4.13. Let $x \in A$. Then $f(x) \in ran\ f$ and therefore by Exercise 4.29, the definition of $g : B \to A$ with Lemma 4.24(4), and (4.8),

$$g \circ f(x) = g(f(x)) = f^{-1}(f(x)) = I_A(x).$$

□

Remark 4.48. Due to Exercise 4.41, $f : A \to B$ is surjective if there exists a right inverse of $f : A \to B$. The converse that there exists a right inverse of $f : A \to B$ if $f : A \to B$ is surjective is true as well. However, to prove this converse, we need an extra axiom. Therefore we will prove it later.

Theorem 4.49. *Let $f : A \to B$ and $g : B \to C$ be functions. Then*

1. *If f and g are injective, then $g \circ f$ is injective*
2. *If f and g are surjective, then $g \circ f$ is surjective*
3. *If f and g are bijective, then $g \circ f$ is bijective.*

Proof. First recall that $g \circ f : A \to C$ is a function due to Theorem 4.27.

1. Let $(x_1, y) \in g \circ f$ and $(x_2, y) \in g \circ f$. Then by the definition of the composition of graphs, there exist z_1 and z_2 such that

$$(x_1, z_1) \in f \quad \wedge \quad (z_1, y) \in g$$

and

$$(x_2, z_2) \in f \quad \wedge \quad (z_2, y) \in g$$

Since g is injective, we obtain $z_1 = z_2$. Thus we have $(x_1, z_1) \in f$ and $(x_2, z_2) = (x_2, z_1) \in f$. Finally, since f is injective, we have $x_1 = x_2$.

2. Let $z \in C$. Then since g is surjective, there exists a $y \in B$ such that $(y, z) \in g$. Moreover, since f is surjective, there exists a $x \in A$ such that $(x, y) \in f$. Finally, by the definition of the composition of graphs,

$$(x, z) \in g \circ f.$$

Therefore, $g \circ f$ is surjective (to C).

3. This follows immediately from (1) and (2).

□

Corollary 4.50. *Let $f : A \to B$ and $g : B \to C$ be invertible functions. Then $g \circ f : A \to C$ is a invertible function.*

Definition 4.51 (Direct image)**.** *Let $f : A \to B$ be a function and C be a subclass of A. The direct image of C under f denoted by $\hat{f}(C)$ is the following subclass of B:*

$$\hat{f}(C) := \{y \in B : \exists x \in C \ni y = f(x)\}.$$

That is, $\hat{f}(C)$ is the class of all the images of elements in C. Moreover, by $\hat{f}$ we denote the graph(?)

$$\hat{f} := \{(C, \hat{f}(C)) : C \subset A\}.$$

Definition 4.52 (Inverse image). *Let $f : A \to B$ be a function and D be a subclass of B. The inverse image of D under f denoted by $\check{f}(D)$ is the following subclass of A:*

$$\check{f}(D) = \{x \in A : \exists y \in D \text{ such that } y = f(x)\}.$$

That is, $\check{f}(D)$ is the class of all the pre-images of elements in D. Moreover, by $\check{f}$ we denote the graph(?)

$$\check{f} := \{(C, \check{f}(C)) : C \subset B\}.$$

Notation 4.53.
1. *Let $a \in A$ and $b \in B$. We write $\hat{f}(a)$ for $\hat{f}(\{a\})$ and $\check{f}(b)$ for $\check{f}(\{b\})$.*
2. *In many books and papers, the simpler notation $f(C)$ and $f^{-1}(D)$ is used instead of $\hat{f}(C)$ and $\check{f}(D)$. However, we keep using notation $\hat{f}(C)$ and $\check{f}(D)$ to avoid confusion with images and pre-images.*

Theorem 4.54. *Let $f : A \to B$ be a function. Then*
1. *If $C \subset A$ and $D \subset A$, then*
$$C = D \implies \hat{f}(C) = \hat{f}(D)$$
2. *If $C \subset B$ and $D \subset B$, then*
$$C = D \implies \check{f}(C) = \check{f}(D).$$

Proof. Because of the similarity of the proof, we only prove (1). Suppose that $C = D$. Then by the axiom of extent,

$$x \in C \iff x \in D$$

Therefore, by the definition of the direct image,

$$\begin{aligned} y \in \hat{f}(C) &\iff \exists x \in C \ni y = f(x) \\ &\iff \exists x \in D \ni y = f(x) \\ &\iff y \in \hat{f}(D). \end{aligned}$$

Finally, by the axiom of extent, $\hat{f}(C) = \hat{f}(D)$. □

Remark 4.55. $\hat{f}(C) = \hat{f}(D)$ does not imply $C = D$. Similarly, $\check{f}(C) = \check{f}(D)$ does not imply $C = D$.

Exercise 4.56. *Let $f : A \to B$ be a function. Then prove or disprove the following statements:*
1. *If $C \subset A$ and $D \subset A$, then*
$$\hat{f}(C) = \hat{f}(D) \implies C = D$$
2. *If $C \subset B$ and $D \subset B$, then*
$$\check{f}(C) = \check{f}(D) \implies C = D.$$

Theorem 4.57. *Let A and B sets and let $f : A \to B$ be a function. Then*
1. *$\hat{f} : \mathcal{P}(A) \to \mathcal{P}(B)$ is a function*
2. *$\check{f} : \mathcal{P}(B) \to \mathcal{P}(A)$ is a function.*

Proof. Because of similarity, we only prove (1). First we claim that $\hat{f}$ is a graph. Recall that

$$\hat{f} := \{(C, \hat{f}(C)) : C \subset A\}.$$

Since A is a set, any subclass C of A is a set by the axiom of subset. Moreover, since B is a set and $\hat{f}(C) \subset B$ for any subclass C of A, $\hat{f}(C)$ is a set. Thus $(C, \hat{f}(C))$ is a set for any subclass C of A. In other words, $\hat{f}$ is a graph. Next we prove that *dom* $\hat{f} = \mathcal{P}(A)$ and *ran* $\hat{f} \subset \mathcal{P}(B)$. Since A is a set, its power set $\mathcal{P}(A)$ is well-defined. Thus by the definition of the power set,

$$\hat{f} = \{(C, \hat{f}(C)) : C \in \mathcal{P}(A)\}. \tag{4.9}$$

Moreover, by the definition of the direct image, for each $C \in \mathcal{P}(A)$, $\hat{f}(C)$ is a subclass of B. In other owrds, $\hat{f}(C) \in \mathcal{P}(B)$ for all $C \in \mathcal{P}(A)$. Therefore from (4.9), it is obvious that *dom* $\hat{f} = \mathcal{P}(A)$ and *ran* $\hat{f} \subset \mathcal{P}(B)$. Finally, it only remains to show that $\hat{f}$ satisfies (F2). Let $(x, y_1) \in \hat{f}$ and $(x, y_2) \in \hat{f}$. Then by (4.9), there exist $C_1 \in \mathcal{P}(A)$ and $C_2 \in \mathcal{P}(A)$ such that

$$(x, y_1) = (C_1, \hat{f}(C_1)) \quad \text{and} \quad (x, y_2) = (C_2, \hat{f}(C_2))$$

By the property of the ordered-pair, $C_1 = x = C_2$, $y_1 = \hat{f}(C_1)$, and $y_2 = \hat{f}(C_2)$. In particular, $C_1 = C_2$ and it implies

$$\hat{f}(C_1) = \hat{f}(C_2)$$

due to Theorem 4.54(1). Therefore, $y_1 = y_2$. □

Theorem 4.58. *Let $f : A \to B$ be a function and $\{D_i\}_{i \in I}$ be a family of subclasses of B. Then*

(1)

$$\check{f}\left(\bigcup_{i \in I} D_i\right) = \bigcup_{i \in I} \check{f}\left(D_i\right)$$

(2)

$$\check{f}\left(\bigcap_{i \in I} D_i\right) = \bigcap_{i \in I} \check{f}\left(D_i\right).$$

Proof. (1) By the definition of the inverse image and the union,

$$\begin{aligned} x \in \check{f}\left(\bigcup_{i \in I} D_i\right) &\iff \exists y \in \bigcup_{i \in I} D_i \ni y = f(x) \\ &\iff \exists i_0 \in I \ni (y \in D_{i_0} \ni y = f(x)) \\ &\iff \exists i_0 \in I \ni x \in \check{f}(D_{i_0}) \\ &\iff x \in \bigcup_{i \in I} \check{f}\left(D_i\right). \end{aligned}$$

(2) By the definition of the inverse image and the intersection,

$$\begin{aligned}
x \in \check{f}\left(\bigcap_{i\in I} D_i\right) &\iff \exists y \in \bigcap_{i\in I} D_i \ni y = f(x) \\
&\iff y \in \bigcap_{i\in I} D_i \quad \text{and} \quad y = f(x) \\
&\iff y \in D_i \text{ for all } i \in I \quad \text{and} \quad y = f(x) \\
&\iff y \in D_i \quad \text{and} \quad y = f(x) \text{ for all } i \in I \\
&\iff \exists y \in D_i \ni y = f(x) \text{ for all } i \in I \\
&\iff x \in \check{f}(D_i) \text{ for all } i \in I \\
&\iff x \in \bigcap_{i\in D_i} \check{f}(D_i).
\end{aligned}$$

The only vague part above is

$$\begin{aligned}
&\exists y \in D_i \ni y = f(x) \text{ for all } i \in I \\
\iff &\ x \in \check{f}(D_i) \text{ for all } i \in I.
\end{aligned}$$

It is obvious that

$$\begin{aligned}
&\exists y \in D_i \ni y = f(x) \text{ for all } i \in I \\
\implies &\ x \in \check{f}(D_i) \text{ for all } i \in I.
\end{aligned}$$

Conversely, assume that

$$x \in \check{f}(D_i) \text{ for all } i \in I.$$

Then for each $i \in I$, there exists $y_i \in D_i$ such that $f(x) = y_i$. However, due to (F2), $y_i = y_j$ for all $i, j \in I$. Therefore, there exists a y such that $f(x) = y$ and $y \in D_i$ for all $i \in I$.

□

Theorem 4.59. *Let $f : A \to B$ be a function and $\{C_i\}_{i\in I}$ be a family of subclasses of A. Then*

(1)

$$\hat{f}\left(\bigcup_{i\in I} C_i\right) = \bigcup_{i\in I} \hat{f}(C_i).$$

(2)

$$\hat{f}\left(\bigcap_{i\in I} C_i\right) \subset \bigcap_{i\in I} \hat{f}(C_i).$$

Proof. (1) By the definition of the direct image and the union,

$$\begin{aligned} y \in \hat{f}\left(\bigcup_{i\in I} C_i\right) &\iff \exists x \in \bigcup_{i\in I} C_i \ni y = f(x) \\ &\iff \exists i_0 \in I \ni (x \in C_{i_0} \ni y = f(x)) \\ &\iff \exists i_0 \in I \ni y \in \hat{f}(C_{i_0}) \\ &\iff y \in \bigcup_{i\in I} \hat{f}\left(C_i\right). \end{aligned}$$

(2) By the definition of the direct image and the intersection,

$$\begin{aligned} y \in \hat{f}\left(\bigcap_{i\in I} C_i\right) &\iff \exists x \in \bigcap_{i\in I} C_i \ni y = f(x) \\ &\iff x \in \bigcap_{i\in I} C_i \quad \text{and} \quad y = f(x) \\ &\iff x \in C_i \text{ for all } i \in I \quad \text{and} \quad y = f(x) \\ &\iff x \in C_i \quad \text{and} \quad y = f(x) \text{ for all } i \in I \\ &\iff \exists x \in C_i \ni y = f(x) \text{ for all } i \in I \\ &\iff y \in \hat{f}(C_i) \text{ for all } i \in I \\ &\iff y \in \bigcap_{i\in C_i} \hat{f}(C_i). \end{aligned}$$

□

Exercise 4.60. *Find an error in the proof of Theorem 4.59(2). Moreover, find an additional condition on $f : A \to B$ to make it correct.*

Exercise 4.61. *Let $f : A \to B$ be a function and $\{C_i\}_{i\in I}$ be a family of subclasses of A. Prove or disprove that*

$$\hat{f}\left(\bigcap_{i\in I} C_i\right) = \bigcap_{i\in I} \hat{f}\left(C_i\right).$$

Definition 4.62 (Choice function)**.** *Let A be a set and write $\mathcal{P}'(A)$ for $\mathcal{P}(A) - \{\emptyset\}$. By a "choice function" for A, we mean a function $r : \mathcal{P}'(A) \to A$ such that*

$$\forall B \in \mathcal{P}'(A), \qquad r(B) \in B.$$

Remark 4.63. Let A be a set and $r : \mathcal{P}'(A) \to A$ be a choice function. Then for each $B \in \mathcal{P}'(A)$, we have $r(B) \in B$. In other words, for each nonempty subset B, $r(B)$ is an element of B. Thus intuitively, we can understand a choice function $r : \mathcal{P}'(A) \to A$ as the action that we choose an element from each nonempty subset of A.

Axiom 4.64 (Axiom of choice)**.** *Every set has a choice function.*

Theorem 4.65. *Let $f : A \to B$ be a function and assume that $A \neq \emptyset$. Then $f : A \to B$ is surjective if and only if there exists a right inverse of $f : A \to B$, i.e. there exists a function $g : B \to A$ such that $f \circ g = I_B$.*

Proof. We skip the proof of the if part due to Exercise 4.41. We only prove the only if part. Assume that $f : A \to B$ is surjective. Then for each $y \in B$, $\check{f}(y) := \check{f}(\{y\})$ is nonempty. Due to the axiom of choice, there exists a choice function r for A. Define

$$g = \{\big(y, r(\check{f}(y))\big) : y \in B\}.$$

Then since $r : \mathcal{P}'(A) \to A$ is a function, $g : B \to A$ is a function. Due to Theorem 4.13, it is sufficient to show that

$$f \circ g(y) = y \qquad \forall y \in B. \tag{4.10}$$

Observe that for each $y \in B$, $g(y) = r(\check{f}(y)) \in \check{f}(y)$. Thus $f(g(y)) = y$. Finally by Exercise 4.29, we have (4.10).

□

Exercise 4.66. *Let $f : A \to B$ be a surjective function and r be a choice function for A. Define*

$$g = \{\big(y, r(\check{f}(y))\big) : y \in B\}.$$

Prove that $g : B \to A$ is a function.

Definition 4.67 (Product of a family of classes)**.** *Let $\{A_i\}_{i\in I}$ be an indexed family of classes and define*

$$A = \bigcup_{i\in I} A_i.$$

The "product" of the classes A_i is defined to be the class

$$\prod_{i\in I} A_i = \{f : \quad f : I \to A \text{ is a function, and } f(i) \in A_i,\ \forall i \in I\}.$$

Example 4.68. *Let $I = \{1, 2\}$, $A_1 = \{a, b\}$, and $A_2 = \{c, d\}$. Then*

$$A = \bigcup_{i\in I} A_i = \{a, b, c, d\}$$

and $\prod_{i\in I} A_i$ consists of all graphs (functions) $f : \{1, 2\} \to \{a, b, c, d\}$ such that $f(1) \in A_1$ and $f(2) \in A_2$. More precisely, there are four functions: $f_1 : \{1, 2\} \to \{a, b, c, d\}$, $f_2 : \{1, 2\} \to \{a, b, c, d\}$, $f_3 : \{1, 2\} \to \{a, b, c, d\}$, $f_4 : \{1, 2\} \to \{a, b, c, d\}$, where

$$f_1 = \{(1, a), (2, c)\}, \quad f_2 = \{(1, a), (2, d)\}, \quad f_3 = \{(1, b), (2, c)\}, \quad f_4 = \{(1, b), (2, d)\}.$$

Therefore,

$$\prod_{i\in I} A_i = \{f_1, f_2, f_3, f_4\}.$$

On the other hand, recall the Cartesian product $A_1 \times A_2$. Then

$$A_1 \times A_2 = \{(a, c), (a, d), (b, c), (b, d)\}.$$

By identifying,

$$\begin{aligned} f_1 &= (a, c),\\ f_2 &= (a, d),\\ f_3 &= (b, c),\\ f_4 &= (b, d), \end{aligned}$$

we may think that $\prod_{i\in I} A_i$ *and* $A_1 \times A_2$ *are equal. In particular, we can easily find a bijective function from* $\prod_{i\in I} A_i$ *to* $A_1 \times A_2$*. In other words,* $\prod_{i\in I} A_i$ *and* $A_1 \times A_2$ *are in one-to-one correspondence.*

Exercise 4.69. *Let* $I = \{1, 2\}$*,* $A_1 = \{a, b\}$*, and* $A_2 = \{c, d\}$*. Find a bijective function from* $\prod_{i\in I} A_i$ *to* $A_1 \times A_2$*.*

Notation 4.70. *Let* $\{A_i\}_{i\in I}$ *be an indexed family of classes and define*

$$A = \bigcup_{i\in I} A_i.$$

- *Let* a *be an element of* $\prod_{i\in I} A_i$*. Then* a *is an function from* I *to* A*. We agree that* a_j *and* $a(j)$ *have the same meaning, i.e.* a_j *is the image of* $j \in I$*. We call* a_j *the* j*-th coordinate of* a*.*
- *Assume that for each* $i \in I$*,* $x_i \in A_i$*. We use the symbol* $\{x_i\}_{i\in I}$ *to designate the element in* $\prod_{i\in I} A_i$ *whose* i*-th coordinate is* x_i*.*
- *Let* $j \in I$*. For all* $a \in \prod_{i\in I} A_i$*, we define* p_j *as*

$$p_j(a) = a_j,$$

i.e. p_j *is the graph given by* $\{(a, a_j) : a \in \prod_{i\in I} A_i\}$*. Then* p_j *becomes a function from* $\prod_{i\in I} A_i$ *to* A_j*. We call this function* $p_j : \prod_{i\in I} A_i \to A_j$ *the "*j*-th projection of* $\prod_{i\in I} A_i$ *to* A_j*.*

Exercise 4.71. *Let* $j \in I$*. Prove that* $p_j : \prod_{i\in I} A_i \to A_j$ *is a function.*

Notation 4.72. *Let* A *and* B *be classes. The symbol* B^A *refers to the class of all functions from* A *to* B*. In particular, if* 2 *denotes a class of two elements, then* 2^A *denotes the class of all functions from* A *to* 2*.*

Theorem 4.73. *Let* A *be a set and* 2 *be a class consisting of two (different) elements* 0 *and* 1*. Then* 2^A *and* $\mathcal{P}(A)$ *are in one-to-one correspondence.*

Proof. It is sufficient to find a bijective function $\gamma : \mathcal{P}(A) \to 2^A$. Recall Example 4.20. For any $B \in \mathcal{P}(A)$, we can consider the characteristic function C_B. More precisely, $C_B : A \to 2$ is a function such that

$$C_B(x) = \begin{cases} 1 & \text{if} \quad x \in B \\ 0 & \text{if} \quad x \notin B \text{ and } x \in A. \end{cases}$$

That for each $B \in \mathcal{P}(A)$, it is obvious that $C_B : A \to 2 \in 2^A$. We define γ by

$$\gamma = \{(B, C_B : A \to 2) : B \in \mathcal{P}(A)\}.$$

Then $\gamma : \mathcal{P}(A) \to 2^A$ is a function such that

$$\gamma(B) = C_B : A \to 2 \qquad \forall B \in \mathcal{P}(A).$$

It only remains to show that the function $\gamma : \mathcal{P}(A) \to 2^A$ is injective and surjective. First we prove that it is injective. Assume that $\gamma(B) = \gamma(D)$ for $B, D \in \mathcal{P}(A)$. Then $C_B : A \to 2 = C_D : A \to 2$. In particular,

$$\begin{aligned} &\{(x, 1) : x \in B\} \cup \{(x, 0) : x \notin B \text{ and } x \in A\} \\ &= C_B \\ &= C_D \\ &= \{(x, 1) : x \in D\} \cup \{(x, 0) : x \notin D \text{ and } x \in A\} \end{aligned} \tag{4.11}$$

Thus by Theorem 3.54,

$$\{(x,1) : x \in B\} = \{(x,1) : x \in D\}. \tag{4.12}$$

Therefore, by Theorem 3.54 again, we have $B = D$. Next we prove that the surjective property. Let $f : A \to 2 \in 2^A$. Put $B = \check{f}(1)$. Then

$$f = \{(x,1) : x \in B\} \cup \{(x,0) : x \notin B \text{ and } x \in A\}$$

and $B \in \mathcal{P}(A)$. Thus

$$f = C_B \text{ with } B \in \mathcal{P}(A).$$

Therefore, $\gamma(B) = f : A \to 2$. □

Exercise 4.74. *Prove that* (4.11) *implies* (4.12).

Remark 4.75. Due to Theorem 4.73, we can identify 2^A and $\mathcal{P}(A)$ in a certain sense. It is the reason why the symbol 2^A is widely used to denote the power set of A in many books and articles of mathematics.

Axiom 4.76 (Axiom of replacement)**.** *If A is a set and $f : A \to B$ is a surjective function, then B is a set.*

Theorem 4.77. *Let $\{A_i\}_{i\in I}$ be an indexed family of sets. Assume that I is a set. Then $\{A_i : i \in I\}$ is a set.*

Proof. Define

$$\phi = \{(i, A_i) : i \in I\}.$$

Then $\phi : I \to \{A_i : i \in I\}$ is a surjective function. Therefore, by the axiom of replacement, $\{A_i : i \in I\}$ is a set. □

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Department of Mathematics, Korea University, 1 Anam-dong, Sungbuk-gu, Seoul, 136-701, Republic of Korea

Email address: waldoo@korea.ac.kr